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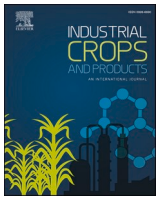

Emerging materials and pretreatment strategies in anaerobic digestion for biogas production: A review on recent advances, limitations, and future perspectives

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ABSTRACT

Anaerobic co-digestion (AcoD) and reliable pretreatment methods are highly beneficial to valorizing waste into valuable biogas production because of the balanced substrate stoichiometry and sufficient recalcitrance. In this regard, materials science is illuminating contemporary research to facilitate effective biomass conversion and biogas generation. The application of materials science in the bioenergy industry is believed to enhance biogas yield significantly while minimizing detrimental effects on the environment and the economy. In this review article, the mechanisms and typical properties of different materials, including MXene, carbon quantum dots (CQDs), graphene, nanomaterials, biocarbon, and iron conductive materials (CMs), were thoroughly investigated. A diverse array of ecological and economic perspectives in the biogas sector were elucidated. Given the potential of case studies in comprehending the current situation, a detailed discussion of anaerobic digestion (AD) and biogas production worldwide was highlighted. The latest advancements in pretreatment methods—biological hydrolysis, steam explosion, and microaeration—were discussed. A comparative bibliographic study over the last decade is more beneficial, and this review briefly examined the frequency of documents in relation to the sustainable development goals (SDGs). Finally, the key research gaps and advancements in the AD and biogas field through machine learning (ML) models were emphasized.

1. Introduction

Biogas production is gaining popularity due to its lower initial and ongoing expenses and its ability to utilize organic waste as valuable precursors (Bianco et al., 2021). Resource depletion and environmental degradation pose significant challenges to achieving global sustainable development goals (SDGs). Using organic waste in sustainable consumption reduces greenhouse gas (GHG) emissions and further supports SDGs 12 (responsible consumption and production) and 13 (climate action). Biomethane production reduces landfill usage and methane emissions by turning organic waste into energy (SDG 15) (Sharma et al., 2023; Guan et al., 2024). Globally, there was an increase in the annual production of solid waste, with 2 gigatonnes in 2020, and by 2050, it is projected to increase to 3.5 gigatonnes, where 46% was credited to organic waste (Dong et al., 2023; Chavan et al., 2022). Therefore, future sustainable bioenergy production will be majorly relying on the different waste sources due to broad availability, cost-effectiveness, and valorization.

In this context, for achieving SDGs in the energy field, researchers and industries must consider key characteristics to improve the AD efficiency: (i) anaerobic co-digestion (AcoD), (ii) pretreatment, and (iii) materials. Unfortunately, the mono-digestion of easily biodegradable substrates like food wastes suffers from typical constraints such as rapid degradation, low C/N ratio, limited buffering capacity, and trace nutritional deficiencies. The rapid decomposition of organic materials leads to an overabundance of volatile fatty acids (VFAs), creating a low pH environment that inhibits microbial growth and insufficient AD. For promoting effective microbial growth, an optimal balance must be maintained between alkaline pH and the proportions of VFAs (Leung and Wang, 2016). Specifically, AcoD can enhance biogas yield by introducing lipids and carbohydrates into the feedstock (Shrestha et al., 2023; Harun et al., 2019).

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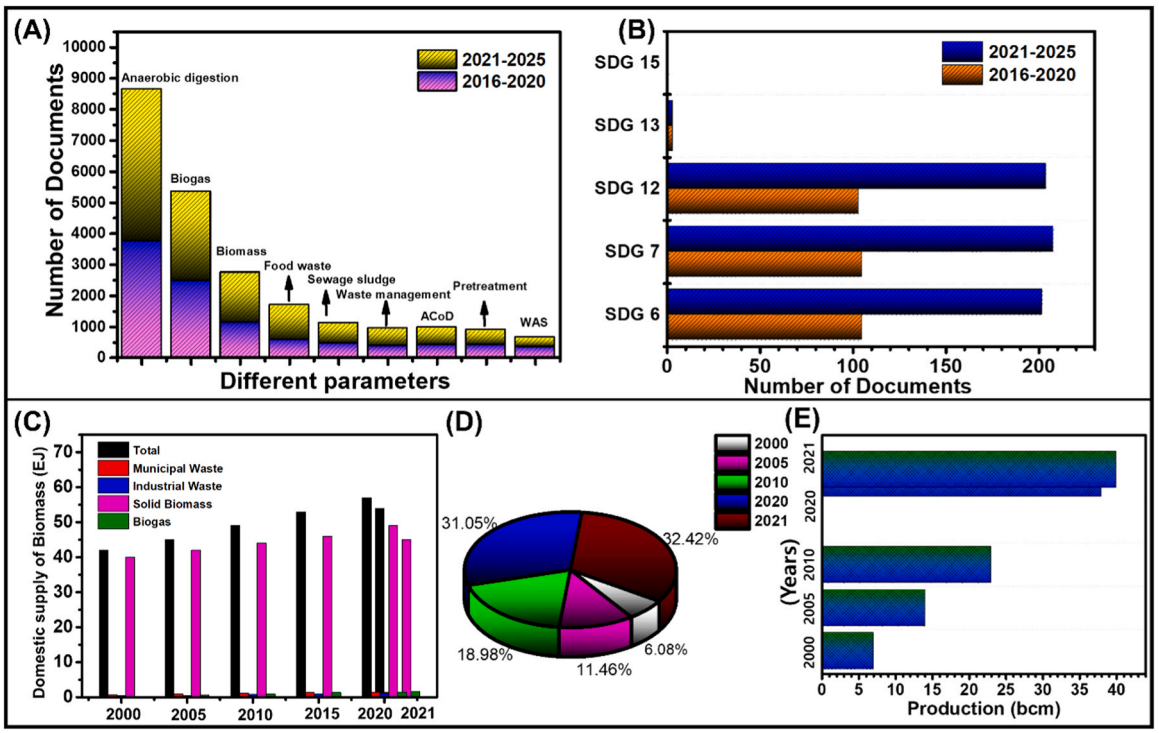

Fig. 1. (A) Chronological data for the frequency of documents and different parameters in the last decade (2016–2025) using Scopus. (B) Bibliometric study regarding the frequency of documents dealing with leading SDGs (6,7,12,13, and 15) in the biogas sector using Web of Science Core Collection. (C) Domestic supply of different biomass from 2000 to 2021. (D, E) Trends in global biogas production.

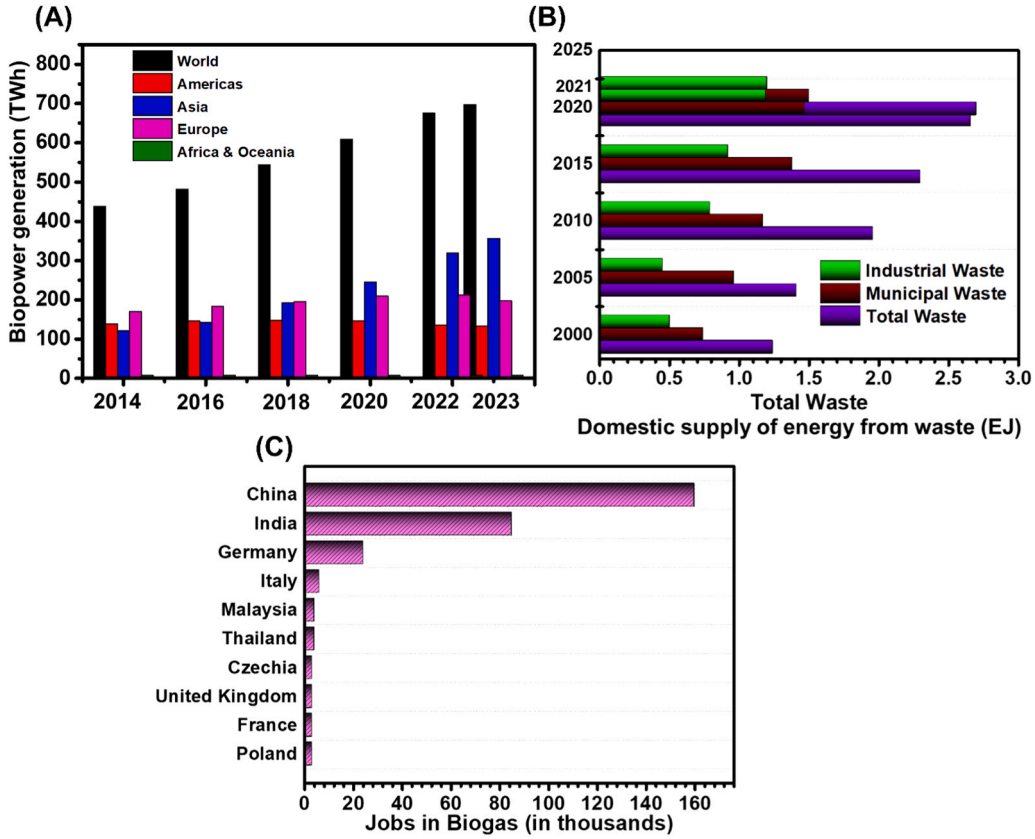

Fig. 2. (A) Global biopower generation from 2014 to 2023. (B) Domestic supply of energy from different wastes from 2000 to 2025. (C) Top 10 countries with the highest employment in biogas production (2022).

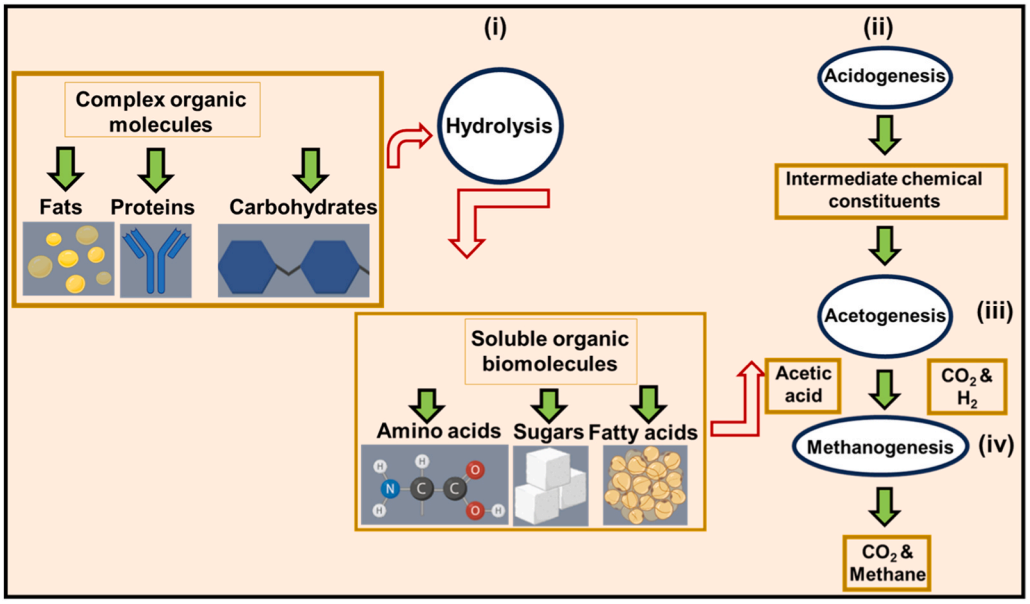

Fig. 3. Different stages involved in the AD process; (i) hydrolysis, (ii) acidogenesis, (iii) acetogenesis, and (iv) methanogenesis.

In the past decade, research studies have shown that incorporating food waste into different organic wastes, such as animal manure, agricultural residue, sewage sludge, and industrial organic wastes, yielded substantial biogas (Zhang et al., 2015; Chen et al., 2020; Amha et al., 2017; Lin et al., 2011). Osei et al. utilized a combination of human excreta (78.8%), food leftovers (11.8%), and kitchen residues (9.4%), which were anaerobically digested to produce high methane yield (764.79 mL CH_4/gVS) (Osei-Owusu et al., 2024). Therefore, multiple combinations provide the greatest synergistic impact (Kanellos et al., 2024).

Based on the above discussion, the chronological data over the last decade (2016–2025) was examined using prominent keywords—"anaerobic digestion," "biogas," and "pretreatment,"—from Scopus (Fig. 1A). Between 2016 and 2020, the frequency of articles for standard parameters was as follows: "AD" (3744), "Biogas" (2469), "Biomass" (1122), "Food waste" (579), "Sewage sludge" (466), "Waste management" (381), "ACoD" (418), "Pretreatment" (414), and "Waste activated sludge (WAS)" (351). Comparatively, in the past five years (2021–2025), significant progress was observed in the frequency of documents: "AD" (4894), "Biogas" (2882), "Biomass" (1618), "Food waste" (1114), "Sewage sludge" (648), "Waste management" (567), "ACoD" (554), "Pretreatment" (477), and "WAS" (303).

Furthermore, a comparable bibliometric study was conducted to understand the research trends of SDGs in the biogas sector. Based on the Web of Science Core Collection, the study was investigated using the keywords "anaerobic digestion," "biogas," and "pretreatment" (Fig. 1B). A total of 122 research publications from 2016 to 2020 were identified as follows: SDG 6: clean water and sanitation (105), SDG 7: affordable and clean energy (105), SDG 12: (103), SDG 13: (3), and SDG 15: life on land (1). From 2021–2025, there was a significant increase in the frequency of documents, with a total of 242 research publications. SDG 7 (208); SDG 12 (204); SDG 6 (202); SDG 13 (3); and SDG 15 (1). These data resources strongly emphasize that the research and academics sector has increasingly concentrated on the biogas sector, primarily aiming to attain significant SDGs.

Therefore, a thorough economic and scientific knowledge must be needed for the utilization, distribution, and valorization of the biomass.

- In 2021, the worldwide domestic supply of biomass approached 54 exajoules. The solid biomass contributes 85% of the domestic supply, and the remainder includes municipal and industrial waste (2–3%) and biogas (2%) sectors (Fig. 1C). In 2021, about 40 billion cubic meters of biogas were yielded worldwide (Fig. 1D, E). Biopower generation has augmented consistently from 439-terawatt hour (TWh) in 2014–697 TWh in 2023, with Asia at the forefront of this expansion. Over the decade, Asia's proportion of global biopower (357 TWh) increased from 28% to 51%, thus exceeding Europe. While the Americas had a slight reduction from 146 TWh to 133 TWh, Africa and Oceania averaged around 7–8 TWh. In 2023, China produced 25% of the global biopower (204 TWh). Noteworthy, the third greatest provider was Japan (18% induced growth rate from 2022), contributing 49 TWh (Fig. 2A).
- In 2021, the domestic energy supply derived from municipal and industrial waste was 2.7 exajoules, with 56% originating from municipal waste and the remainder from industrial waste (Fig. 2B). Based on these considerations, the biogas industry had substantial employment creation in 2023, propelled by the establishment of more than 120 plants in the USA and over 300 in Europe. The biogas sector generated approximately 316,000 jobs worldwide. Especially China, which dominated with almost 50% of these positions, although India, bolstered by biogas-centric policy, contributed 85,000 jobs. Among European Union countries, Germany was the main employer, generating 24,000 jobs (Fig. 2C) (Global Statistics Report, 2024).

In the last two decades, various nanomaterials have been responsible for improving the AD by optimizing the pH and enzyme activities of e^- transporters. The integration of diversified nanoscale materials such as nano-Al_2O_3, nano-Fe_2O_3, and carbon nanotubes (CNTs) has led to enhanced biogas and methane yields (Zhao et al., 2023a, 2023b; Yang et al., 2012). Especially, these nanomaterials can efficiently biodegrade VFAs and induce direct interspecies electron transfer (DIET) in divergent microbial species during AD. Moreover, CNTs are accountable for enhanced methanogenesis due to high negative redox potential, dose-dependent cytotoxic effects, and even non-DIET mechanisms (Ziganshina et al., 2022; Salvador et al., 2017). Despite offering high surface area, high electrical conductivity, and stability features, CNTs face significant challenges regarding their high expensiveness, environmental concerns, and insufficient scalability for large-scale biogas production.

In this regard, understanding how divergent materials generate biogas is crucial in this context, and till now, earlier reports were

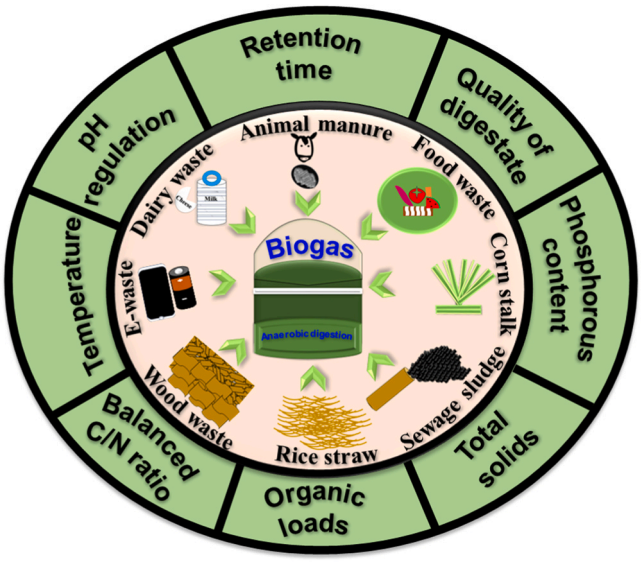

Fig. 4. Schematic representation for the key characteristics for the enhanced AD process and miscellaneous substrates.

devoted to the progress of individual materials in AD and biogas production. Unfortunately, to the best of the author's knowledge, there were no reports emphasizing the broad range of materials with this concept. Hence, in this review article, a thorough focus was given to the various mechanisms, limitations, and advancements of the materials in the biogas field.

2. Significance of anaerobic co-digestion

Generally, AD involves four major biochemical processes, such as hydrolysis, acidogenesis, acetogenesis, and methanogenesis stages (Fig. 3). For the effective AD process, proper co-digestion of miscellaneous substrates contributes to a favorable environment for heterogenous microbial growth. During AcoD, these substrates maintain optimal pH to enhance biogas yield because of their intrinsic buffering capacity and distinguished chemical composition. For instance, Zhou and his colleagues determined how the optimal pH range (5.25–11) affects the VFAs during AD of Chinese cabbage waste. Based on the substrate breakdown rates, VFA conversion, and microbial community assessment, at an ideal pH of 6.0, 30.4% of VFAs were produced during AD, and a high methane yield was achieved (Zhou et al., 2021).

- A pilot-scale case study conducted at a hawker facility in Singapore demonstrated that decentralized AD of food waste for 71 days leads to a remarkable difference in TS%, VS%, and C/N ratios (Tiong et al., 2023). The study has yielded higher biogas with greater potential in contrast to the pilot systems in Bolivia, the UK, and Italy. The main contribution factors include (i) utilization of an energy-saving AD system with a digester volume; (ii) climatic conditions; the tropical climate of Singapore (around 29°C); (iii) nutrient-rich food waste digestates (P, K, and N); and (iv) an ideal total inorganic carbon value (0.3–0.4).

Various case studies on AD efficiency were examined, along with environmental and economic aspects, as follows:

- A case study has been conducted in New South Wales, Australia, and has evaluated the effects of co-digestion for the feasible production of biomethane via agricultural waste biomass. From economic and ecological evaluations, the lowest levelized cost of energy (LCOE) (A $19.7/GJ) and the efficient CO_2 reduction performance (-58.9 kt CO_2-e/year) were identified. The biomass pricing exerts the most significant influence, and the charge of A$40/t for biomass acquisition would significantly decrease the LCOE of biomethane production (A$11.3/GJ), rendering the price of biomethane cheaper than the average natural gas price (Hosseini et al., 2024).
- The ranching industry emits abundant GHGs and contaminates both air and water sources. In this context, AD methods for the treatment of animal manure offer a cost-effective valorization approach via energy conservation, condensed digestate production, and power generation. A preliminary study of 123 pig farms in Taiwan showed that the farms produced 3.7×10^7 kWh of electricity, mitigated methane emissions by 6.1 gigagrams per year, saved USD 4.0 million per year on electricity costs, and reduced CO_2 emissions by 152.5 kilotonnes per year (Tsai, 2018).

The temperature and pH extremes are adversely affecting the AD efficiency through causing VFA accumulation and bacterial suppression. For example, thermophilic (high temperature) conditions are better for lignocellulose-based substrates when it comes to biochemical reaction rates, retention periods, and biogas yields than mesophilic (low temperature) conditions. The digestate quality, total solid (TS), phosphorus, and pH must be satisfied for the convenient AD (Fig. 4). It is also critical to consider the efficient pretreatment strategies for the successful AD. Recently, a number of studies have also shown that biological hydrolysis, microaerobic pretreatment, and steam explosion are the most useful pretreatment methods.

Researchers have developed thermal, mechanical, biological, and chemical disintegrations in recent decades to enhance biosolid hydrolysis and AD. To consider the better pretreatment strategy in improving the biogas yield, some typical parameters need to be optimized. These include (i) SCOD, (ii) soluble proteins, (iii) ammonia nitrogen, and (iv) VFA concentration (Ünşar et al., 2018; El Gnaoui et al., 2022).

3. Advances in pretreatment strategies

A proper balance between pH and VFA accumulation during the AD process is crucial for achieving desirable biogas yield. Although earlier methods, such as physical, chemical, mechanical, and biological pretreatments, have been utilized for the degradation of complex organic structures on different substrates, a single method alone is not sufficient for controlling the operating conditions. Therefore, combining two or more pretreatments in a tailor-made approach could benefit maximum biogas production with less energy needed, reduce carbon emissions, and mitigate environmental pollution.

The selection of solid substances is crucial for efficient pretreatment because each material has its advantages and limitations. For example, carbon-based solids such as biochar, activated carbon, graphene, and MXenes are derived from pyrolysis, exfoliation, or chemical activation (Wang et al., 2021a, 2021b, 2021c; Ponzelli et al., 2024). Therefore, these materials display different characteristics: high porosity, surface functionality, and conductivity, and serving as microbial carriers, facilitate DIET, and adsorb inhibitory intermediates, ultimately enhancing methane yield and stabilizing digestion performance (Xiao et al., 2021; Bahrami et al., 2025). Coming to the mineral and alkaline solids, including lime, dolomite, MgO, zeolites, and clays, they originate from natural mineral deposits or calcination processes. These solids are predominantly enriching buffering capacity, maintaining neutral pH, and adsorbing ammonium and heavy metals. In this way, these materials are inclusively beneficial in feedstocks with high nitrogen content, such as sewage sludge and food waste (Fischer et al., 2025; Bahmanzadegan and Ghaemi, 2025; Baloch et al., 2023). Metal- and oxide-based solids, such as zero-valent iron, magnetite, and hematite, are generally synthesized by the oxidation or reduction of iron salts. The redox mediating properties catalyze electron transfer and consequently induce enzymatic activity (Jin et al., 2025; Lee et al., 2023).

Recently, composite materials, such as Fe–biochar, Ni–biochar, and Fe_3O_4–graphene, integrate multiple functionalities into a single additive.

These materials provide better synergistic effects by enhancing electrical conductivity, adsorption, and catalytic properties compared with a single additive (Li et al., 2024a; Wang et al., 2021a–c). Overall performance of these materials depends on the type of feedstock and pretreatment method for improving AD efficiency. The above discussion signifies the rationale for investigating a range of materials rather than focusing on a single additive. Furthermore, the ecological and economic aspects of these materials also play an important role. For instance, biochar, produced from waste biomass, reduces GHG emissions and improves digestate quality when applied to soils. Although zeolites and clays are cost-effective and abundant in nature, their performance is inferior compared to other materials. On the other hand, MXenes and CQDs seem to be promising, but they still face scalability and cost challenges (Liu et al., 2023a, 2023b, 2023c, 2023d; Viaene et al., 2024; Zheng et al., 2025).

Overall, the above discussion on solid substances signifies the role of material diversification in AD. Future research should emphasize composite materials rather than relying on a single material that exploits synergistic relationships among them. These strategies will enhance and stabilize microbial consortia and hydrolysis, promote DIET, and ultimately achieve more sustainable methane yields.

3.1. Physical pretreatment

This process is mainly used for the conversion of lignocellulose biomass from different substrates, including food waste and others into fractional components (cellulose, hemicellulose, and lignin). For instance, in the extrusion process, by employing heat and mechanical shear through the extruder, physical characteristics of the substrates can be altered, which include increased surface area, reduced particle size, and its reactivity. The other physical pretreatments include grinding, ultrasound, and ball milling, etc.

Extrusion runs at mild temperature, with few chemicals and moderate pH conditions, and causes limited sugar degradation through avoiding the formation of harsh toxic compounds during pretreatment. A rigorous literature is available for this process utilizing agricultural wastes such as corn stover, and under specific conditions, woody substrates such as hardwood, eucalyptus, and wood chips were also employed (Karunanithy and Muthukumarappan, 2010; Zhang et al., 2012; Chen et al., 2013; Lin and Ryu, 2014; Senturk-Ozer et al., 2011; Lee et al., 2009, 2010; Karunanithy et al., 2012). From a thorough research literature investigation, it is advisable that still lab and full-scale sectors are facing challenges when pretreatment is exclusively dependent on the extrusion process due to the several limitations.

In this context, the smart integration of extrusion pretreatment with miscellaneous pretreatments such as alkaline extrusion, acid hydrolysis, steam explosion, ammonia fiber explosion, microwave treatment, vacuum extrusion, and bioextrusion is seems to be promising. Over the last four decades, several patents were efficiently issued for the extrusion alone, and combining the extrusion with other pretreatments (Huber et al., 1988; Stirl et al., 2008; Vilarem et al., 2015).

Moreover, milling pretreatment is a highly mechanical mode of physical pretreatment, which uses knife mills, hammer mills, and ball mills for degrading different substrates into micro-sized particles. It is ideally promising for reducing particle sizes in agricultural wastes such as wheat straw and corn stover. However, it has exerted certain limitations similar to the extrusion pretreatment, and additionally, others include (i) the formation of flammable dust during ultra-fine milling of substrates, which requires explosion-proof gadgets; (ii) limited scalability for industrial purposes due to economic burden; (iii) high limitation for wet biomass such as food waste and sludge-based feedstocks; (iv) severe degradability in heat-sensitive compounds like sugars; and (v) rapid damage to the blades during the pretreatment of hard biomass like wood-based substrates. Although ultrasound, autoclaving, and pulsed electric field (PEF) pretreatments exist, significant operational and scale-up limitations persist.

3.2. Chemical pretreatment

Recently, in the field of "green chemistry," scientists around the world have been working on eco-friendly ways to treat biomass before it is used. The goal is to potentially utilize the biomass, consume less waste, and avoid using harsh and harmful chemicals.

- Deep eutectic solvents (DESs) are one of the best advanced green pretreatment methods for biomass owing to its non-purification synthesis approach, reduced waste, minimal toxicity, and rapid degradation in the environment. Liang and his colleagues studied lactic acid-derived DESs with hydrogen bond acceptors for the breakdown of maize (*Zea mays* L.) stover with minimal cost-burden and hydrolyzing the polysaccharide into fermentable sugars efficiently (Liang et al., 2021a).
- Achieving desirable energy conversion efficiency by utilizing DES-based pretreatment is highly significant for producing biomethane. Almost 80% of the lignin was extracted from the willow biomass by pretreating at160°C for15 min lactic acid-derived DESs (choline chloride to lactic acid; molar ratio of 1:10). Because of this, the digestion time was minimized from 22 to 10 days, and a high biomethane yield (89.9 mL/g TS-122.5 mL/g TS) was achieved, and the high energy conversion efficiency of 62.7% suggests that DES pretreatment is a potential and sustainable approach (Kang et al., 2023).
- Recently, an ammonium thiocyanate: urea-based DES pretreatment has substantially increased the biomethane yield (48%) of maize stover (Olugbemide et al., 2021). Furthermore, it's important to examine other DESs for treating different substrates in biomethane production.
- The choline chloride and oxalic acid in a 1:1 ratio forms a DES, where oxalic acid primarily influencing the duration of the lag phase, while choline chloride enhances biogas yield and supplies abundant carbon for the anaerobic consortium. After the completion of the lag phase, the DES at optimal concentration of 12.5 g/L has demonstrated the increased biogas yield (Lima et al., 2021).

3.3. Physicochemical pretreatment

3.3.1. Steam explosion

The steam explosion (SE) pretreatment is an excellent thermochemical pretreatment method because it is quick, effective at breaking down water, and uses high-pressure steam. In this approach, pressurized steam is introduced into a reactor containing biomass for a specified pressure and residence time. The steam penetrates the cellulosic structure, and a sudden release of pressure causes explosive decompression, leading to the efficient breakdown of the biomass. It is often triggered at temperatures (160–260°C) and pressures (0.69–4.83 MPa) within a few minutes, before the substrate is subjected to steam. The high temperatures can efficiently break down plant-based waste, which constitutes cellulose, into hemicellulose and lignin, which makes the microbial AD easier to initiate (Vochozka et al., 2016; Hoang et al., 2023).

It is widely known that most of the agricultural residues, including sugarcane bagasse, straw, and fibers from flax and hemp, possess lesser lignin and cellulose contents. The hemicellulose backbone of these residues primarily consists of β-1,4-linked arabinoxylan or arabinose-glucuronoxylan polymers, with branched side chains of arabinose and glucuronic acid. During SE pretreatment, branched side chains are hydrolyzed into monomeric sugars, including arabinose, glucuronic acid, and xylose. Likewise, SE pretreatment is more convenient for processing agricultural feedstocks than for wood-based substrates, which contain abundant recalcitrant lignin that complicates the ability of SE pretreatment to sufficiently disrupt their structures.

It is crucial to understand the surface morphology of the substrates before and after the pretreatment operation. Through this analysis, researchers can assess effective structural changes in the substrates and mitigate the recalcitrance of biomass. Tupciauskas et al. have

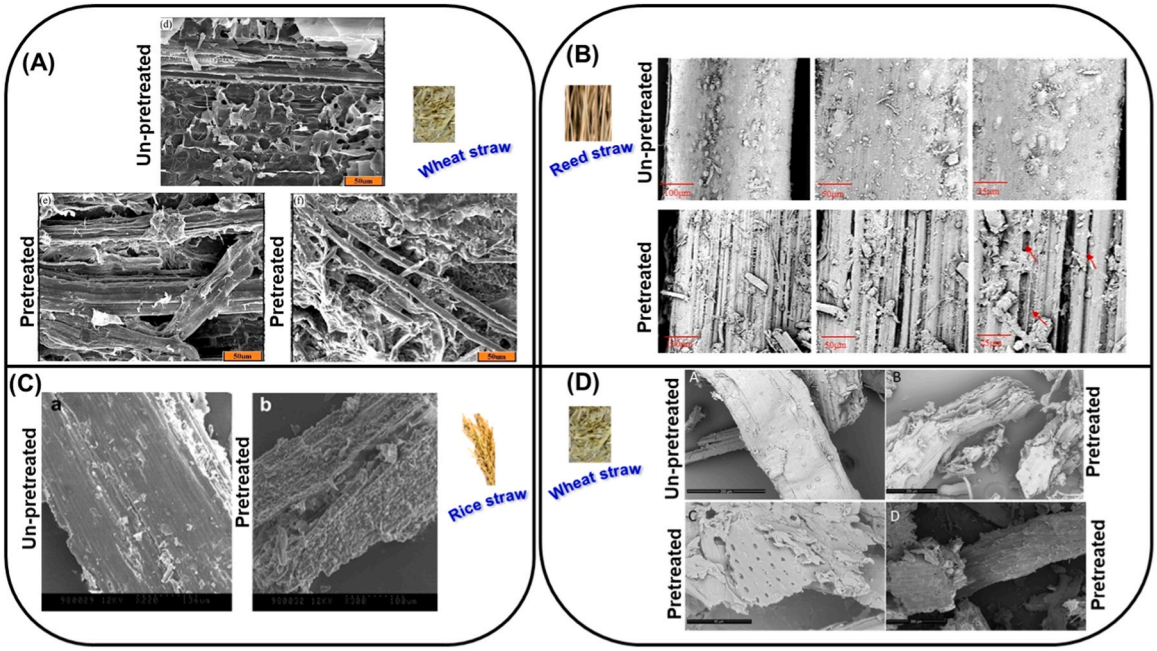

Fig. 5. Morphological features of different substrates before and after pretreatment. SEM micrographs of (A) un-pretreated wheat straw and pre-treated at SE200/1 and SE240/3. (B) reed straw before and after SE pretreatment. (C) rice straw before and after SCE pretreatment. (D) un-pretreated wheat straw and SCE pre-treatment at 300 bar, 100°C, PFD, and 300 bar, 150°C. Adapted from (Tupciauskas et al., 2021; Liu et al., 2025; Llorens et al., 2024; Gao et al., 2010).

investigated the morphological characteristics, such as the particle size reduction of the pre-treated wheat straw, by employing the scanning electron microscopy (SEM) technique at two different operating conditions, such as SE200/1 (200°C, 1.0 min) and SE240/3 (240°C, 3.0 min). However, wheat straw is significantly affected by SE pre-treatment due to its thinner epidermis layer, and further, its natural structure was slightly disrupted after the pre-treatment at SE200/1 to the fibre bundles and single fibres. Specifically, the increase in SE pretreatment intensifies the disruption of cell structure in wheat straw. On the other hand, SEM images of the wheat straw pre-treated at SE240/3 have displayed severely disrupted smaller dense layered fibre bundles of melted lignin in the form of spherical droplets measuring < 5 μm over the surfaces of few fibres(Fig. 5A) (Tupciauskas et al., 2021).

The surface of the unpretreated reed straw was uniformly arranged with intact fiber structures, and after SE pretreatment (170°C, 2.5 MPa) for 5.0 min, peculiar alternations in the morphology of reed straw were identified. The disrupted surfaces of the pretreated reed straw were significantly remarkable with highly exposed cellulose vascular bundles, enlarged pores, and conspicuous cracks, which eventually result in the increased surface area in contrast to the unpretreated reed straw (Fig. 5B) (Liu et al., 2025).

The SE pretreatment contributes several benefits as compared with that of contemporary pretreatments, which include minimal environmental effect, fewer chemicals, and significant energy efficiency (Avellar and Glasser, 1998). More importantly, many traditional mechanical approaches are compromised by utilizing approximately two-thirds more energy in contrast to the SE pretreatment, and thus, it is quite convenient and significant for saving operational cost burden.

The latest advancements in different pretreatment strategies associated with AD and biogas yield were discussed as follows:

- A weak acid-driven SE can effectively break down xylan. Proton (H^+) ions from H_2SO_4 can effectively penetrate the cell wall of rice straw (RS) during an SE, and break down the hemicellulose and defibrillate the RS. A pilot-scale (240 kg/day) SE of RS was conducted using diluted acid to yield abundant fermentable sugars, which could demonstrate bigger economic and energy benefits during the scaling-up process (Semwal et al., 2024).
- Currently, many white-rot fungi, like *Pleurotus ostreatus* and *Lentinula edodes*, are edible mushrooms that act as valorizing agents during the pretreatment. Hence, by digesting these agents with biomass, industrial biogas plants can produce efficient biogas and can reduce investment expenses.
- The combination of Fenton's reagent and fungi with the pretreated SE birch wood (SEBW) was utilized to improve the rate and yield of methane during the AD. Here, researchers determined the optimal conditions for Fenton's reagent by maximizing the binding of ferric (Fe^{3+}) and ferrous (Fe^{2+}) iron to the SEBW substrate. The SE pretreatment has significantly boosted the binding of Fe^{2+} to the wood structure. The ideal doses of Fenton's reagents and fungal pretreatments were found to be 1 mM Fe^{3+} and 0.5 mM Fe^{2+}, which resulted in high methane yields (352 and 383 mL/g VS) after 40 days (Hashemi et al., 2022).

3.3.2. *Supercritical carbon dioxide explosion*

In the supercritical CO_2 explosion (SCE) method, substrates are being rapidly compressed using supercritical CO_2 ($scCO_2$) under a specific time period, which leads to the enhanced degradability. It is a conveniently sustainable and greener approach due to the utilization of easily vaporized CO_2 gas resulting in lower energy consumption with a high biogas yield. Especially, CO_2 is favorable because of its renewability, lower toxicity, and cost-effectiveness, which can work smoothly in critical conditions such as 31.1°C and 7.36 MPa. Putrino et al. employed $scCO_2$ (5 h) for delignification of cellulosic compounds (coconut fibers) under vigorous compressed pressure (70°C, 20 MPa), leading to the breakdown of cellulosic structures (Putrino et al., 2020). The key alterations in the fibers such as increased porosity, limited wax, and minimized phenolic contents, as well as the weakening of hydrogen bonds between cellulose and lignin, are majorly responsible for the efficient delignification. Even more, the surface area will be enhanced and facilitate the frequency of enzymatic hydrolysis in the cellulosic compounds, and consequently, results in the increased glucose content (540.9 μM).

The performance of the SCE pretreatment strongly depends on the type of feedstock. For example, minimal enzymatic hydrolysis with less than a 30% increase in the sugar yield with the pretreatment of rice

husk, corn stover, and RS when compared with that of wheat straw, switchgrass, and sugarcane bagasse, where more than 100% of increased sugar yield was observed (Narayanaswamy, et al., 2011; Gao et al., 2010; Pasquini et al., 2005). In another study, approximately a 34.5% of increase in the sugar yield was noted during the pretreatment of cornstalk, corncob, and RS (Liu et al., 2014). Zhang et al. utilized combinational SCE and ultrasonic pretreatments using sorghum stalk as the feedstock and improved sugar yield (33.69%) by enzyme hydrolysis (Zhang et al., 2019a, 2019b). Overall, SCE is more beneficial as compared with the SE pretreatment and can be conveniently run at a lower temperature without degrading sugars.

Fig. 5. **C** shows unpretreated and pretreated (30 MPa, 110°C, 30 min) SEM images of RS, which revealed obvious morphological changes. Initially, without pretreatment, smoothly compacted and adjoined surfaces were found; however, post-pretreatment results in extensively porous and stalked structures of RS with fluffy fibres. The highly exposed microfibrils facilitate multi-fasciculi-inducing enzyme entry and subsequently, lead to the substantial enzymatic hydrolysis. Therefore, $scCO_2$ effectively disrupts the physical structures of RS and allows microfibrils with increased roughness and surface area, which serves to yield sufficient enzymatic hydrolysis. Previous literature also supports these results, which demonstrated that pretreated RS was much softer to the touch as compared with that of untreated RS (Gao et al., 2010).

Llorens et al. found severe fibre disruption in wheat straw structures after SCE pretreatment under high pressure (Fig. 5D). The pretreated wheat straw at 300 bar shows extensive surface area (53%) when compared with the untreated wheat straw. These findings were highly satisfactory with the earlier reports (Llorens et al., 2024).

These studies are highlighting the significant role of the SCE pretreatment in the efficient utilization of biomass like rice and wheat straw. In future studies, it is advisable to investigate the pretreatment at higher temperatures for promoting the successful biomass conversion.

3.4. Biological pretreatment

For instance, Gnaoui et al. optimized biogas yield using pretreatments, with maximum generation observed when biological hydrolysis (BH) was employed, starting at 37°C for 20 h and terminating at 55°C for 20 h. In the BH pretreatment, an enhanced methane yield of 674.49 mL CH_4/g VS was observed, attributed to the accumulation of VFAs at 37°C over a 40-hour incubation period. (El Gnaoui et al., 2022; Chen et al., 2021). Thus, sustainable and cost-effective pretreatment strategies can overcome the constraints of the above methods.

For more than fifteen years, the BH pretreatment has been employed in full-scale production of biogas (Bungay and Abdelwahab, 2008). During BH pretreatment, a wide range of enzymes, bacteria, and fungi participate in the depolymerization of extracellular polymeric substances (EPS) and microbial cell walls within the temperature range of 35–70°C. Its main advantages include (i) cost-effective downstream processing, (ii) minimal formation of inhibitory compounds, and (iii) lower energy consumption (Thapa et al., 2024).

- BH pretreatment has improved biosolid reduction, COD, and VFA changes in AD. Adding calcium peroxide (CaO_2) and chilling promoted abundant short-chain fatty acids (SCFAs) with a sufficient biogas yield (438.5 mg COD/g VSS) (Zhao et al., 2023a, 2023b; Gao et al., 2024).
- In a case study of Pakistan, authors compared different pretreatment-inducing agents such as $Ca(OH)_2$, H_2O_2, SiO_2, and *P. ostreatus* for estimating methane yield (Noonari et al., 2023). Burning canola straw, a byproduct of canola seed cultivation, generally causes pollution in Pakistan. Thus, an efficient biological degradation was occurred in canola straw via treating above agents. Maximum methane yields of 417.3, 386.1, 381.0, and 238.6 mL CH_4/g VS were obtained at 0.4% $Ca(OH)_2$, 0.2% H_2O_2, 0.5 mg SiO_2, and *P. ostreatus*, respectively. $Ca(OH)_2$ generates 7.4, 8.7, and 42.8% higher percentage of methane than to other pretreatments.
- This case study contributed the key pros and cons of these agents as follows: (a) The acidic environment of H_2O_2 is lethal to methanogens, which mitigate the methane yield; (b) whereas, the purified SiO_2 affords a cost burden and is not as effective as other NPs; (c) the *P. ostreatus* pretreatment requires a long incubation time and well-optimized conditions for microbial growth; and (d) comparatively, $Ca(OH)_2$ is more potential than these agents due to cost-effectiveness, alkalinity, and rapid allowance of methanogens. Thus, these beneficials made $Ca(OH)_2$ a favorable pretreatment choice.
- Another study from Punjab Agricultural University, India, has employed RS for BH pretreatment with a microbial consortium, consisting of *Paenibaccilus ehimensis, Arthrobacter nicotianae, Pseudomonas* sp., *Delftia* sp., *Aspergillus* sp., *Trichoderma* sp., *Bacillus* sp., and *Lysinibacillus fusiform*. After 10 days of pretreatment with a cellulolytic microbial consortium, the methane yield increased by ~45.9% (187.45 L/kg RS) as compared with the untreated RS with 128.45 L/kg RS of biogas. In this context, lignocellulolytic microorganisms are highly significant for BH pretreatment with high degrading abilities in lignocellulosic biomass (Sahil et al., 2023).

3.4.1. Thermo-biological pretreatment

Although several hydrothermal pretreatments are currently being used in industrial plants for sludge digestion, high energy requirements for elevated temperatures, increased ammonia levels, and the presence of recalcitrant organic matter have negatively impacted anaerobic bacteria during the AD process (Bicakci and Eskicioglu, 2019; Xiong et al., 2023; Cui et al., 2022; Rahmani et al., 2022; Wu et al., 2024). Altogether, these recently observed limitations opened a new avenue for temperature-phased AD (TPAD)-based pretreatment to enhance biogas production in a convenient manner.

For over 15 years, TPAD pretreatment has played a crucial role in producing high methane yields, and it is operated at thermophilic and mesophilic temperatures by facilitating the solubilization of complex substrates and stable operation, respectively. For instance, a study from France produced 40% of methane after 48 h of pretreatment with cattle slurry and maize silage substrates. This study signifies that hydrolysis rates greatly showed the impact on particulate matter and solubilization around the temperatures between 55°C and 65°C (Dooms et al., 2018). Recently, Wu and his colleagues employed TPAD utilizing mesophilic (35°C) and thermophilic (55°C) temperatures and produced higher methane (25%) and high TS removal (10%) with a greater hydrolysis rate (20%) using sewage sludge. Due to the enhanced hydrolysis of the main fraction protein from the sewage sludge, there was a greater increase in loadings from 3.96 g COD/L/d to 8.05 g COD/L/d and a half-fold reduction in hydraulic retention time (HRT) from 20 days to 10 days. The methane yield has improved from 0.11 L/g COD added to 0.15 L/g COD added (Wu et al., 2025). After a thorough literature study, several similar reports were identified using TPAD, facilitating convenient AD on different sludge-based substrates (Gavala et al., 2003; Ge et al., 2010, 2011; Kim et al., 2011; Orozco et al., 2013).

3.4.2. Microaerobic pretreatment

Microaerobic pretreatment can accelerate the AD because a small amount of oxygen facilitates both anaerobic and aerobic biological activities in the same bioreactor with sufficient biogas yield.

- This method mainly benefits facultative microbes (i.e., *Lactobacillus*) by sufficiently enhancing the concentrations of VFA and LA, which leads to a desirable methane yield of 349.1 mL/g VS. For instance, for tobacco straw substrate, at oxygen levels of 4 mL/g VS sufficiently inhibited the growth of microbes and modified its physicochemical properties (Wang et al., 2024a, 2024b).

Table 1
Summary of different pretreatment approaches in the methane or biogas production in the last five years.

S. No.	Substrate	Time duration	Pretreatment	Methane/biogas yield (mL/g VS)	Ref
1.	Chestnut shell	24 h	Hydrothermal–alkaline pretreatment	253 (±9)	(Bianco et al., 2021)
2.	Food waste	40 h	BPH at 37°C	674.49	(Gnaoui et al., 2022)
3.	Steam-exploded birch wood (SEBW)	40 days	Fenton's reaction pretreatment	383 [0.5 mM Fe(II)] and 352 [1.0 mM Fe(III)]	(Hashemi et al., 2022)
4.	SEBW	40 days	Fungal pretreatment	420 (Pretreated with *Lentinula edodes*) and 419 (Pretreated with *Pleurotus ostreatus and* 1.0 mM Fe(III)+H_2O_2)	(Hashemi et al., 2022)
5.	Canola straw	40 days	Chemical pretreatment [0.4% $Ca(OH)_2$, 0.2% H_2O_2, 0.5 mg SiO_2, and *P. ostreatus* (for 30 days)]	417.3, 386.1, 381.0, and 238.6	(Noonari et al., 2023)
6.	Willow (*Salix purpurea* L.)	10 days	Chemical pretreatment [DESs (choline chloride-lactic acid (ChCl:LA)]	122.5 mL/g TS	(Kang et al., 2023)
7.	Rice straw	45 days	Physicochemical pretreatment (DESs)	467.84	(Basak et al., 2022)
8.	Cork dust	130°C for 60 min	Chemical pretreatment (DESs: choline chloride and formic acid (1:2 M ratio)	115.1 mL N gVS^{-1}	(Elmaci et al., 2023)
9.	Cork dust	130°C for 60 min	Chemical pretreatment (DESs: choline chloride and formic acid (1:2 M ratio)	525 ± 3 mL N gVS^{-1}	(Schultz et al., 2025)
10.	Food waste	30 min	Thermal pretreatment at 100°C	806.1	(Gnaoui et al., 2022)
11.	Giant grass (Anaerobic) and cow manure (Microaerobic)	25 days	Anaerobic and Microaerobic pretreatment (O_2 concentration: 10 mL/g VS)	275.3, 279.8, and 265.3	(Wang et al., 2021)
12.	Tobacco straw	50 days	Microaerobic pretreatment (O_2 concentration: 4 mL/g VS)	349.1	(Wang et al., 2024)

- Using this method, compost made from giant grass straw inoculated with cow manure at a concentration of 10 mL/g VS led to higher methane yields (Wang et al., 2021a, 2021b, 2021c; Ramos and Fdz-Polanco, 2013).
- Lim et al. studied the effects of microaeration pretreatment for 4 days by co-digestion of brown water and food waste, using 37.5 mL O_2/L/day, which produced 10–21% higher methane yield. Efficient solubilization and acidification of the pretreated inoculated sludge could achieve a high yield. Due to the microaeration pretreatment, the inoculum could have boosted the enzymatic response of hydrolytic and acidogenic bacteria (Lim et al., 2013). In another study, this method has driven higher hydrolysis rates through AD of sewage sludge, sufficiently accounted for increased SCOD (65.14%), and outstanding methane yield (221%) in 40 h. The highest microbial activities of 597%, 170%, and 79.4% were observed for 20, 30, and 40 h of pretreatment (Rashvanlou et al., 2020). Noteworthy, increased SCOD and microbial activities observed from this pretreatment are efficient as compared with those from thermal, chemical, and mechanical pretreatments (Li et al., 2022a; Le et al., 2013).

Over the past decade, researchers have focused on the microaeration technique, leading to significant advancements in this area of study, as follows:

- In general, adding a small amount of oxygen to an anaerobic fermenter helps sulfide-oxidizing bacteria grow, and mainly the superior electronegativity of the oxygen triggers the electron-accepting performance, which transforms hydrogen sulfide (H_2S) into elemental sulfur. Even though microaeration is often used in large-scale applications, intermittently stirred fermenters still have unprecedented challenges that could make microaeration ineffective, specifically during the upgrading of biogas into biomethane. Thus, it is more important to maintain optimal H_2S levels around 5.0 ppm in the substrates comprising higher sulfur contents during AD. For instance, certain woods (wood chips from camphor), sewage sludge, and other agricultural residues (apple leaves and winter wheat straw) mainly consist of higher sulfur proportions.
- Andreides et al. optimized the various operational parameters, where initial stirring speed (15 RPM for 2 h) followed by an increased stirring strength to 35 RPM for an additional 2 h resulted in the highest amount of H_2S in biogas dropping from 6000 to < 40 ppm (Andreides et al., 2021). Moreover, post-pretreatment processes can further remove the additional concentrations of H_2S during AD. This study has achieved an outstanding H_2S removal efficiency (97%), which is nearly correlated with other reported studies (Ramos et al., 2014; Jenícek et al., 2017; Huertas et al., 2020; Shu et al., 2025). Thus, the intermittent stirring offered crucial merits such as sufficient VFA production, enhanced microbial population, efficient H_2S removal, and biogas upgradation.
- Oxidation reduction potential-based micro-aeration has been studied for the AD of lignocellulosic biomass at elevated organic loading rates (OLRs). A study conducted by Nguyen et al. maintained the intermittent ORP at −470 mV (i.e., +25 mV above the anaerobic baseline ORP) and achieved excellent VFA inhibition (Nguyen et al., 2019). Especially, O_2 is mediating as the limiting electron acceptor for facultative heterotrophs, facilitating partial aerobic oxidation of

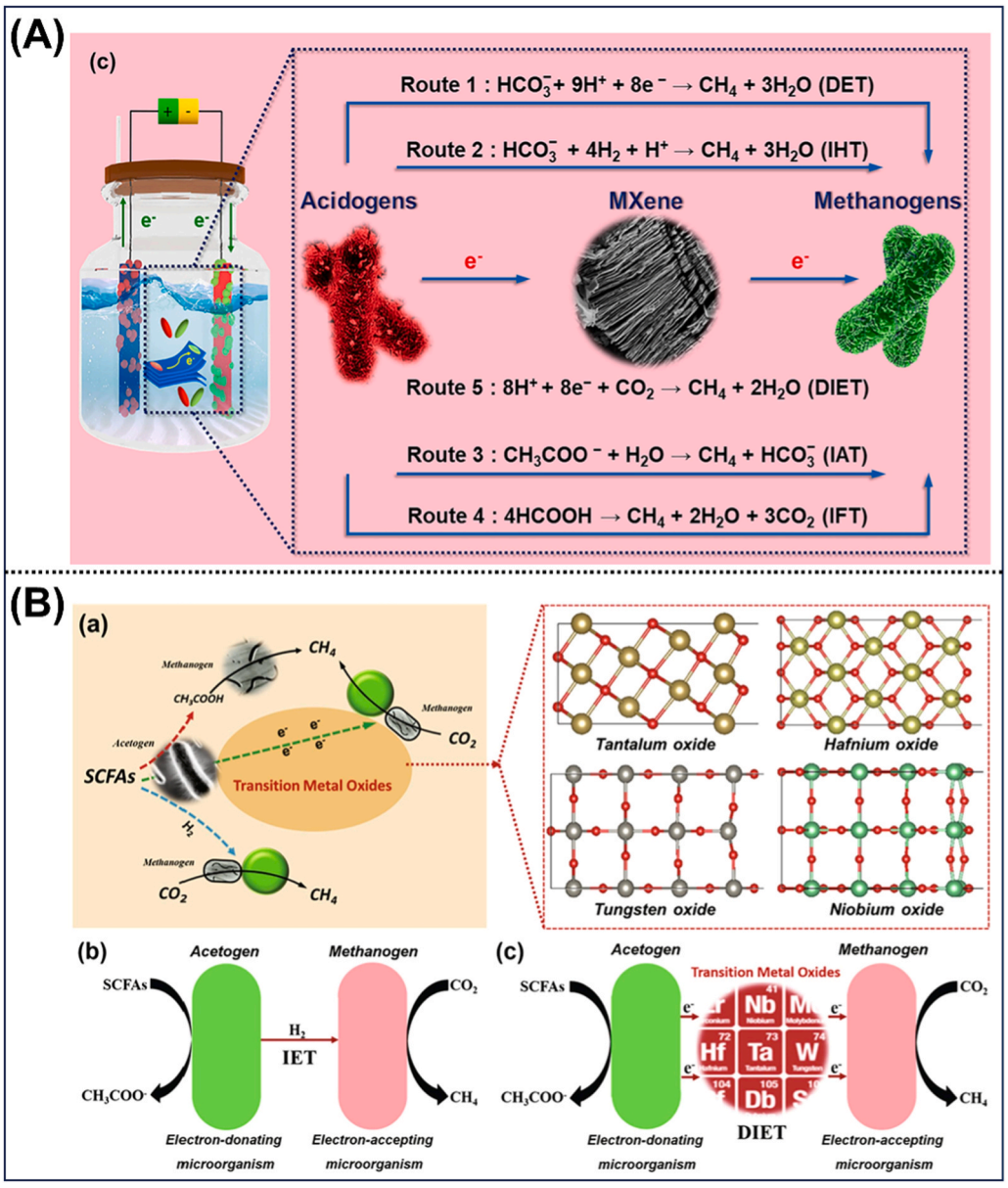

Fig. 6. (A) Schematic illustration of the working mechanism of MEC-AD system with ML-MXene accelerants. (B) (a) Mechanism of biogas production in AD systems, and (b) the IET, and (c) the enhanced DIET by TMOs-based accelerants. Adapted from (Liu et al., 2023; Yun et al., 2021).

VFA while preserving essential anaerobic environments for methanogens.

- Wu et al. employed micro-aeration using intermittent ORP levels (-470, −450, and −475 mV) with a dissolved oxygen concentration of 0.62 pM. Authors identified the methanogen *Proteiniphilum* as an effective cellulolytic facultative bacterium for the degradation of lignocellulose by aerobic and anaerobic fermentation. Significantly, antioxidant enzymes available in *Proteiniphilum* could detoxify oxygen to yield a high methane level of 80.8 ± 9.6 N mL/VSg (Wu et al., 2021). Consequently, this method is being vastly optimized for high-loading AD systems in compromised digesters with extremely accumulated VFAs (Khanal et al., 2003; Khanal and Huang, 2006; Wu et al., 2023). The previously reported pretreatment approaches in the methane or biogas production over the last five years are summarized in Table 1.

4. Emerging materials for the biogas production via AD process

Besides their accelerating effects in improving AD, it is also important to consider the structural derivation of solid substances and their separation after use so that processing conditions and implications can be thoroughly explored for practical application.

Carbon-based solids, such as biochar and activated carbon, are typically derived from the pyrolysis or activation of biomass residues such as wood, agricultural waste, and sewage sludge. Their micro- and mesoporous architecture, enriched surface functional groups, and graphitic domains result from the thermal or chemical treatment conditions. These structures facilitate microbial colonization, electron shuttling, and adsorption of inhibitors during AD (Ngatiman et al., 2025; Li et al., 2025). However, once employed, these materials generally remain embedded within the digestate matrix and recovery could be possible under magnetic stimulation, and predominantly, digestate-derived biochar is often used as a soil conditioner to aid in secondary valorization (Mahmood et al., 2025). In general, mineral and

alkaline solids are either naturally mined or produced by calcination. Their crystalline structures, ion-exchange properties, and buffering capacity contribute to pH stabilization and ammonium/heavy metal adsorption during digestion. While partly soluble (as in the case of lime and MgO), these substances typically persist as insoluble residues in the digestate. They are rarely separated but rather integrated into biosolids, improving their agronomic quality (Bahmanzadegan and Ghaemi, 2025; Luo et al., 2025; Osman et al., 2022).

Metal- and oxide-based solids, such as zero-valent iron, magnetite, and hematite, are synthesized through methods like reduction of salts, co-precipitation, or mechanical milling. The nanoscale morphology and high surface reactivity of these materials enable catalytic redox reactions, H_2S inhibition, and stimulation of DIET. Depending on their redox nature, these materials ionize into specific ionic species (Fe^{2+}/Fe^{3+}) or remain as particulate residues. Magnetic oxides, i.e., Fe_3O_4, can be separated and reused through magnetic recovery, improving cost-efficiency and reducing the environmental burden of metals (Li et al., 2022b, 2023; Zheng et al., 2022; Zhu et al., 2025). Meanwhile, composite solids such as Fe–biochar, Ni–biochar, and Fe_3O_4–graphene are synthesized by impregnating carbon matrices with metal oxides or NPs, or by assembling carbon–inorganic nanohybrids. These materials sufficiently tune multifunctional features such as high conductivity, catalytic activity, and adsorption capacity, and they serve as highly effective accelerants (Li et al., 2024b; Wang et al., 2025a,b; Zhao et al., 2025). Noteworthy, their engineered composition allows for easier recovery: for instance, magnetic biochar can be separated from digestate using external magnets, enabling multiple reuse cycles and reducing secondary pollution.

Taken together, the structural derivation and fate of these solids highlight their potential and practical challenges. Although carbon-based materials and mineral additives are sustainable and cost-effective, their residual presence in digestates makes valorization majorly dependent on agricultural recycling. Metallic and hybrid composites provide stronger catalytic effects and can be efficiently recovered; however, they are expensive and release NPs into the environment. Therefore, a comprehensive understanding of the accelerating effects, structural derivation, and recovery strategies of miscellaneous solid substances is crucial for evaluating practical applicability. By thoroughly studying the above phenomena, more sustainable and efficient pretreatment strategies can be developed, which maximize methane yields via ensuring environmental safety and economic feasibility.

4.1. Transition metal carbonitrides and oxides

Accelerants are chemical compounds that are added to AD systems to help break down organic wastes faster, encourage microbial activity, increase biogas yield, and sustain stability. For example, fungi, microbial consortia, enzymes, chemical reagents, macronutrients, minerals, trace elements, transition metal compounds, and carbon materials are acting as suitable accelerants. Till now, there isn't much evidence to support the DIET mechanism linked to transition metal-based accelerants in AD. For instance, MXenes are porous carbides with the general formula M_2C, where M stands for Ti, Hf, V, Nb, Cr, Zr, Ta, W, and Mo, and are promising at upgrading biogas through the effective separation of CO_2 from CH_4.

To get insightful understanding, DFT simulation studies have suggested that MXenes can absorb and selectively trap CO_2 via strong interactions between unmodified MXenes and CO_2. Ti_3C_2 MXenes mimic porous, sulfone-functionalized graphene oxide (GO) materials, which can absorb 12 mol CO_2/kg. Comparatively, MXenes are selected better than porous graphene but less than sulfone-functionalized GO. It is noteworthy to mention that CO_2 selectivity and adsorption energies rely on the metal's d electron count in the MXene. From DFT study simulations, MXenes' biogas segregation has increased, as evidentially based on thermodynamic and statistical thermodynamic rate constants. Especially, Grimme D3 dispersion correction in the DFT simulation studies demonstrates that MXenes are conveniently selective at high temperatures and low CO_2 levels.

Furthermore, based on the calculations from adsorption and desorption rate constants for MXenes, i.e., Hf_2C and W_2C, it is confirmed that these MXenes are highly selective for separating CO_2 from CH_4 at room temperature (Prats et al., 2020).

- The multilayer MXene (ML-MXene) has played a crucial role in promoting the growth of divergent bacterial phyla such as *Proteobacteria*, *Bacteroidetes*, and *Firmicutes*. Consequently, ML-MXene can digest organic wastes, COD, and complex chemical constituents and consumes abundant VFAs in the microbial electrolysis cell coupled-AD (MEC-AD) system (Anas et al., 2021; Bellucci et al., 2020; Wang et al., 2022).
- Lv and his research group also discovered that *Methanofollis* solely generates methane from H_2/CO_2 and formate utilizing the ML-MXene (Lv et al., 2020). The ML-MXene can boost the methanogenesis through extracellular electron transfer (EET) using electrochemically active microbial catalysts (Fig. 6A). *Synergisita* and *Methanosarcina* are the common bacteria used in MEC-AD systems, where ML-MXene has facilitated the production of C-type cytochrome and electrically conductive pili (e-pili) and subsequently increased the DIET methanogenesis (Liu et al., 2023a, 2023b, 2023c, 2023d; Van Eerten-Jansen et al., 2015; Lienemann et al., 2018).

In recent times, researchers explored the role of various transition metal oxides (TMOs) such as tantalum oxide (TaO_x), niobium oxide (Nb_2O_5), hafnium oxide (HfO_2), and tungsten oxide (WO_2) as accelerants in AD systems. The insights of TMO-based accelerants were discussed as follows:

- AD performance was investigated to improve the biogas yield using TMOs-based accelerants via divergent parameters: (i) pH, (ii) COD, (iii) TS, (iv) VS, (v) stability, and (vi) fertility of the digestates. In this regard, the DFT calculations were critical for obtaining the theoretical perspectives because of the evidential correlation between TMOs and the DIET mechanism.
- TMOs-based accelerants have higher electron transmission capabilities, which contribute to the improved DIET. From the DFT simulations, the total density of states was close to the Fermi energy level for four surfaces of TMOs. Moreover, TMOs exhibited characteristic metallic properties, with no bandgaps detected, signifying that electrons may readily travel inside these materials to metals.
- TMOs-based accelerants are excellent materials to improve the stability of the AD system via thorough digestion and consumption of organic matter and FAs. A plethora of microbial species utilize SCFAs during the AD by initial transformation of SCFAs into acetate and thereby CO_2, hydrogen, and electrons into methane (Fig. 6B). Hence, TMO-based accelerants are better at inducing the DIET than the IET in AD systems, which can yield more biogas than systems that don't have accelerants (Yun et al., 2021).

The following key parameters need to be considered for the TMO-based accelerants regarding the choice of materials for improving the biogas yield.

(a) Preparing TMO-based composites by mixing TMOs with carbon materials or other MOs seems to be a beneficial way to improve AD performance.
(b) The digestates linked to TMOs had a favorable nutritional composition, and hence, combining TMOs with phosphorus-rich materials is an excellent approach to increase biogas yield.
(c) The economic burden for the TMOs production at lab and industrial facilities varies owing to the different operation modes during the biogas production. Therefore, a thorough monitoring

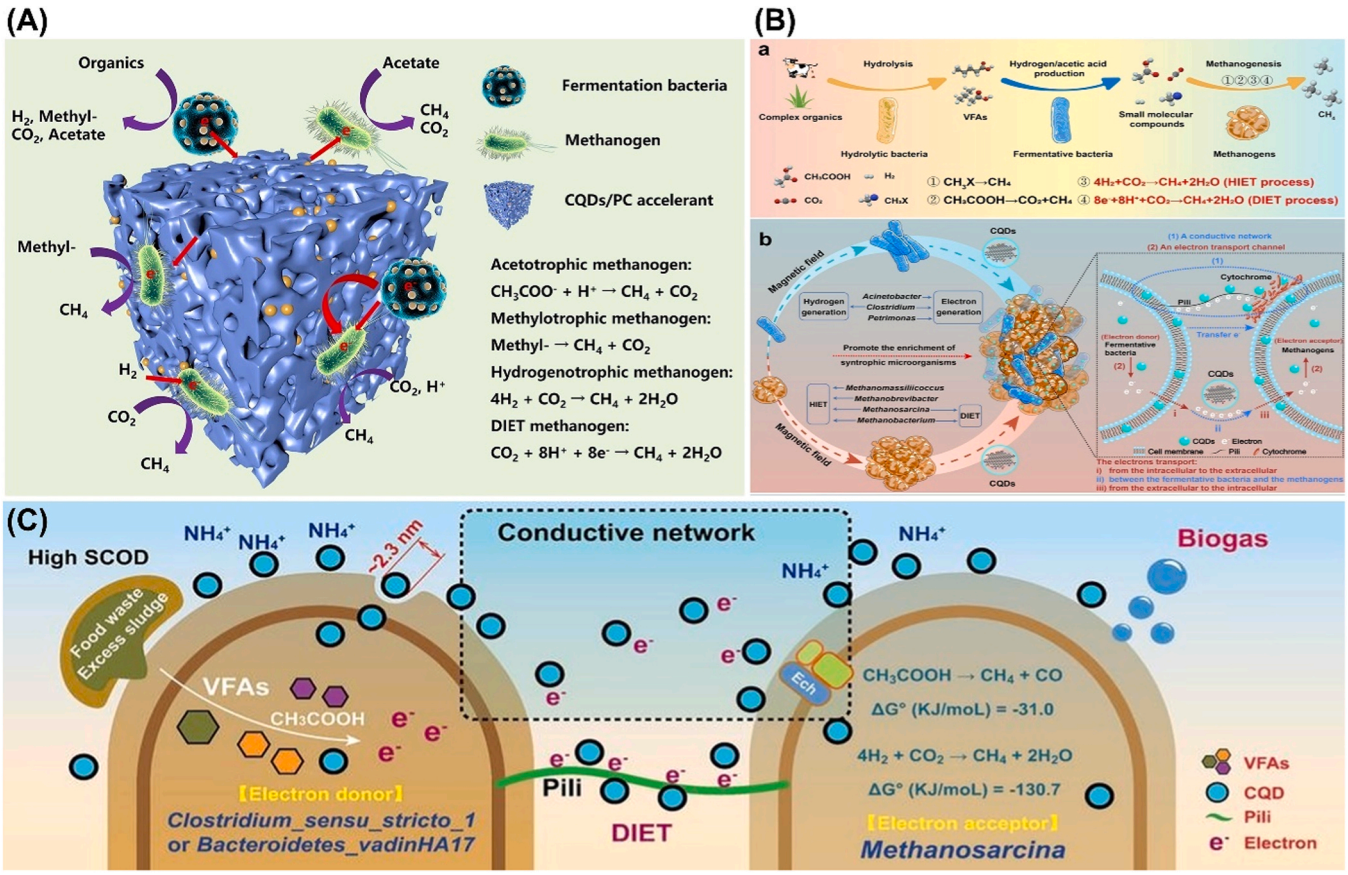

Fig. 7. (A) Schematic diagram of the methane production process of AcoD enhanced by CQDs/PC; (B) Scheme for methanogenesis in AcoD with CQDs under a magnetic field. (C) Schematic diagram of potential mechanism on different CQDs promoting high-load AD. Adapted from (Liu et al., 2023, 2024; Gao et al., 2024).

of optimization and quantification in the production cost to income is urgently needed for both facilities.

(d) TMOs are being detrimental to the ecosystem of the soil, and thus, the removal of these compounds from the digestate is a sophisticated approach to mitigate this risk. As a result, treating TMOs with a magnetic field is an alternative that is easier to extract out of the digestate. Furthermore, a proper visualization of the DIET of TMOs is essential in future research.

4.2. Carbon quantum dots

Carbon quantum dots (CQDs) embedded in porous carbon (PC) work as a host-guest accelerator to improve AcoD under the magnetic field, resulting in TCOD removal and leading to enhanced biogas yield. Significantly, the desired synergistic effect between the improved fertilization effectiveness and CQDs/PC use in DIET together is mainly responsible for enhanced thermal stability of digestate by thorough replenishing of anaerobic bacterial species and finally leads to an efficient AD (Fig. 7A). The following are the detailed perspectives related to the pristine CQDs and CQD-based composites in AD and biogas production:

- The diversified structural microbial community has demonstrated that CQDs/PC composite enhances DIET-associated hydrolytic bacteria and methanogens with the fertilization effectiveness by 4.9%, suggesting the significance of digestate as a biofertilizer (Liu et al., 2023a, 2023b, 2023c, 2023d).
- Further research study suggested that aloe peel and dairy manure-derived AP-CQDs are acting as accelerants, which constitute the promising total nutritional content capacity and recycle resources and result in mitigating COD emissions (Liu et al., 2023a, 2023b, 2023c, 2023d).
- Interestingly, CQDs can considerably improve AcoD energy recovery under magnetic field impact. The synergistic effect between the magnetic field and CQDs resulted in the nurturing of abundant hydrogen-producing bacteria, hydrogenotrophic methanogens, electroactive bacteria, and electrotrophic methanogens.
- Especially, HIET and DIET were induced via ET routes and conductivity networks, which are efficiently controlled by CQDs. The overall combined effects of CQDs and the magnetic field can sustain the energy recovery and IET efficiency in AcoD systems (Fig. 7B).
- In general, e-pili, cytochrome C, membrane-bound redox proteins, natural polymers, and EPS are intracellular electron mediators, which conduct electrons, and form the biofilm matrix, thereby promoting the DIET. The CQDs are biocompatible and form a conductive network in the intracellular, EPS, and digestate portions during AcoD. Furthermore, CQDs can interact with e-pili and cytochrome C of cells via electrostatic interactions and hydrogen bonds, thereby allowing CQDs to serve as potential electrical conduits between microbes in AcoD systems (Gao et al., 2024).
- More importantly, CQDs can enhance strong binding capabilities on microbial surfaces. In this context, the ultra-nanoscale level straw-derived CQDs exhibited a high density of negatively charged organic moieties that can bind with positively charged NH_4^+ ions, thus alleviating the inhibitory effects of ammonia nitrogen on microbial activity (Fig. 7 C) (Liu et al., 2024a, 2024b). The CQDs can be internalized by microbes through endocytosis, further facilitating redox metabolism. These conductive networks functioned as a medium for enhancing electroactive bacteria (i.e., *Clostridium*) and initiating ET. Overall, the enriched microbial surface environment and the ET network significantly improved the AD performance.

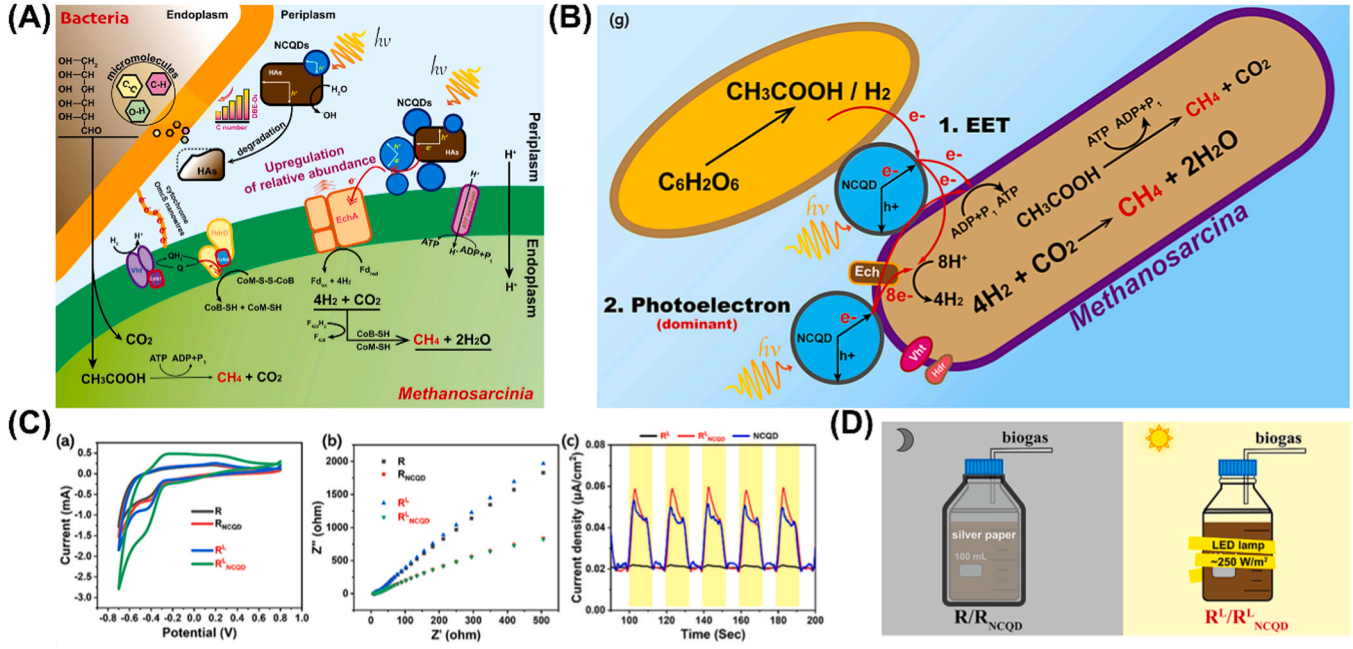

Fig. 8. (A) The mechanism of NCQDs in HAs-inhibited photo-AD systems. The effect of NCQD on methanogenesis in photo-AD system. (B) Schematic diagram of NCQD promoting methanogenesis in photo-AD system. (C) CV curves, and impedances of substrates, and the effect of light on the current in different systems. (D) The schematic diagrams of dark and photo bioreactors. Adapted from (Liu et al., 2024, 2023).

As AD performance is typically dependent on the metabolic activities of the microbial species, it is highly preferential to monitor various adverse factors involved. Among these, humic acids (HAs), VFAs, and induced ammonia nitrogen are leading constraining factors. For instance, the straw-derived CQDs can interact with HAs; photocatalytic effects enable CQDs to generate abundant photoelectrons and oxidative species, ultimately leading to the degradation of HAs. Subsequently, micromolecules can serve as microbial substrates and induce the microbial population and metabolic activities and finally promote the efficient photo-driven methanogenesis (Fig. 8A) (Liu et al., 2024a, 2024b).

Under light treatment, the electrons in NCQDs were excited from the valence to the conduction band, which enabled the methanogenesis through ET between bacteria and archaea. The NCQDs form the biotic-abiotic hybrid interface between the methanogen (*methanosarcina*) and the bacterial cell, and the transferred photoelectrons were utilized for ATP and CH_4 production (Liu et al., 2023a, 2023b, 2023c, 2023d). Thus, the methanogenesis triggered by NCQDs under the visible light is the basic mechanism in the photo-AD system (Fig. 8B).

Liu et al. proposed the mechanistic potentials of NCQDs in the photo-AD system by investigating photoelectrochemical properties. The CV analysis revealed that all curves exhibited redox peaks at 0.22 V and −0.40 V. The DIET process exerted by microorganisms was confirmed through the oxidation peak, and the ET resulted from the reduction peak. However, the NCQDs in dark-AD have not affected the redox peak, and especially, the strength of the reduction peak has been enhanced under light treatment, which suggests that the light mediated the ET pathway. Again, the NCQDs are significantly responsible for the sufficient reduction in impedance in the AD system, whereas the dark conditions mitigated the formation of active species. Furthermore, NCQDs can also supply abundant photoelectrons to microbes, and the I-t curve confirmed that the photocurrent intensity of R^L (photo AD without NCQDs) does not fluctuate in the light/dark cycle, which again showed the lack of photo-response from the microorganisms (Fig. 8C). Remarkably, the photocurrent of the NCQDs has displayed rapid fluctuations in the light and dark cycles due to the significant photoelectrons absorbed by microorganisms. More importantly, the photocurrent intensity of R^L_{NCQDs} (photo-AD with 0.5 g/L NCQDs) has increased greater than that of NCQDs because of the swift absorption of photoelectrons by electron acceptors over the surfaces of microbial cell membranes. This study has used the photo-AD experimental device, i.e., Automatic Methane Potential Test System II (AMPTSII®), in which a 100 ML serum bottle served as the fermentation unit, which was wrapped by silver paper to avoid the light and further wrapped by an LED lamp (20 W) with a simulated light source of total power (250 W/m^2) (Fig. 8D). More importantly, NCQDs in photo-AD did not result in the extreme formation of free radicals, and hence posed no potential threat to microorganisms. These photoelectrochemical studies signify major consequences of NCQDs in the photo-AD system, which include progressed EET and induced photoelectrons for sufficient methanogenesis.

Furthermore, these studies pioneered the key phenomena, such as photoelectrochemical characteristics of substrates, microbial community structure, and enzyme abundance, elucidating the complex mechanisms of light-responsive materials in the photo-AD system. The analysis reveals how variations in substrate properties influence enzyme abundance, ultimately affecting the efficiency of light-responsive materials. By elucidating these complex mechanisms, researchers can better understand the dynamics of microbial interactions and implications for enhancing sustainable energy production through photobiological processes.

Recently, CQDs have demonstrated the potential performance in AD reactors by stabilizing microbial communities under high ammonia nitrogen environments and improving biogas yield. As CQDs function as effective electron mediators, promoting DIET among diverse microbial communities. Through the successful conversion of VFAs, CQDs are improving the resilience of aerobic microbes and elevating ammonia nitrogen levels, subsequently facilitating effective synergistic interactions among various methanogenesis pathways (Qian et al., 2022).

The incorporation of CDs into CO_2 electromethanogenesis devices has considerable promise for practical engineering applications. From a technological standpoint, CDs have significant benefits over other carbon-based CMs. For example, materials like biochar, CNTs, and GO need 10–25 g·L^{-1} to increase methane production by the same amount. On the other hand, 1.0 mL·L^{-1} of CDs increases methane yield by 35.3%. It works better with a lower dose, transports electrons more efficiently with a current density of -19.5 ± 15.1 mA·cm^{-2}, and forms more

Table 2
Overview of miscellaneous materials in the methane or biogas production.

Materials	Substrate	Weight%	TCOD degradation	Methane/biogas yield (mL/g VS)	Ref
Aloe peel-derived CQDs	Cow manure And Aloe peel waste	0.36	58.4%	348.9	(Gao et al., 2024)
Aloe peel-derived CQDs	Daily manure and sewage sludge	0.36	57.87%	570.86	(Liu et al., 2023)
MXene Accelerants	Cattle manure and sewage sludge	0.035	76.6%	358.7	(Liu et al., 2023)
TMOs-based accelerants (TaO_x, Nb_2O_5, HfO_2, and WO_2)	Dairy manure and wastewater sludge	0.01	67.17%–70.45%	565.01–617.85	(Yun et al., 2021)

biofilm with a 91.3% live cell coverage.

As a result of better energy recovery, the energy balance study suggests that the idea might be viable from an economic perspective. For its widespread use, researchers must resolve significant engineering challenges as follows: (i) cost-effective synthesis of CDs is ideal for large-scale production, (ii) preparing automated dosing systems for accurate delivery of CDs in industrial reactors, and (iii) finding new ways for recovering and recycling eco-friendly CDs.

A carbon management system for cities that uses CDs to boost electromethanogenesis could lead to long-term growth. In this way, initial anaerobic treatment of municipal waste water results in a biogas yield consisting of approximately 60% of CO_2. Through using CDs-enhanced microbial electrolysis cells, the biochemically produced CO_2 and the CO_2 that is released by automobiles and factories in cities would both be converted into methane. As generated methane could serve as a power source, generating a cyclic energy system within metropolitan facilities. This approach would concurrently tackle waste water treatment, carbon emission reduction, and renewable energy generation, signifying a substantial progression towards sustainable urban development (Wang et al., 2025a, 2025b). Overall, these applications serve as a basis for the CDs as a sustainable material in developing both economically and environmentally reliable products with less cost burden and operational stability, which can be sufficiently utilized for the large-scale industrial platform in the near future. The overview of various materials for enhancing the methane or biogas yield is listed in Table 2.

4.3. Nanoparticles and other inorganic nanomaterials

Nanotechnology in biofuel production may solve the critical challenges such as feedstock bioconversion, biogas yield, and cost burden. Especially the maximum adsorption, reactivity, and catalytic capabilities of nanoparticles (NPs) can overcome these challenges (Sheikh et al., 2023). For instance, NPs exhibit diverse applications in energy and environmental fields via exceptional properties, which include nano-scale dimensions, high dispersibility, a high surface-to-volume ratio, significant chemical reactivity, extensive surface area, surface charge, and self-assembly. It enables a sophisticated cellular uptake of NPs in methanogens, and further synergistic effects between metabolic and enzymatic activities as well as microbial heterogeneity contributed to the completion of the state-of-the-art AD. Previously, many earlier studies have mentioned the critical constraints of divergent NPs, including Ag, Ni, Fe_2O_3, Al_2O_3, ZnO, TiO_2, and CuO, on various substrates during the AD.

For combating the constraints, this review article primarily emphasized commonly shared mechanisms of nanomaterials/NPs. For instance, metal-based NPs can potentially boost the antibacterial action by promoting the microbial growth via DIET, enzyme interactions, syntrophic associations, and metabolic efficiency. The operational approach involves coating anaerobic sludge with metal-based NPs. When these NPs disintegrate macromolecules in cell membranes, microbes effectively consume sufficient biomass. At extremely low doses (10 ppm), the NPs suppress the synthesis of inhibitory metabolites that might impair the AD.

It is very significant to assess the interactions between metal-based NPs and different enzymes for producing methane. Under low pH conditions, an increase in the activity of metal-based enzymes, i.e., hydrogenase and alcohol dehydrogenase, is mainly accountable for initiating the abiotic dissolution of metal-based NPs and promoting the AD (Wang et al., 2016; Elreedy et al., 2024; Hassanein et al., 2021; Ajay et al., 2020; Rashid et al., 2023). Three primary factors were identified for the promotion of enzymatic activity, which include (i) the convenient metal-dependent enzymatic reactions, (ii) the metal reduction initiated by syntrophic metal-reducing bacteria, and (iii) the increased stability and activity of enzymes through adsorption on NPs.

More importantly, different NPs can stimulate or inhibit AD depending on several factors, including physical and chemical properties, dose or concentration, exposure period, COD removal, and TS and VS% values (Kavitha et al., 2019).

- The physical and chemical properties of rice husk changed when it was exposed to visible light and treated with a nanocomposite of indium vanadate-coated titanium dioxide ($InVO_4/TiO_2$) and yielded more biogas (Rashid et al., 2023).
- Nickel NPs accelerate microalgal biomass breakdown in phase-separated biological pretreatment (PSBP), improving biomethane recovery. The best dose for the Ni NPs was found to be 0.004 g/g of suspended solids, which can weaken cell walls without disintegrating them. Herein, biological pretreatment alone solubilized 24% of the biomass, whereas PSBP solubilized 36%. Biomethanogenesis tests also indicated that PSBP produced 411 mL/g more COD methane (Juntupally et al., 2022; Suanon et al., 2017).
- The understanding of optimal temperature settings is indispensable for achieving desirable biogas yield. Significantly, conductive NPs such as iron, nano zero-valent iron (nZVI), and Fe_3O_4 are highly affecting AD of waste sludge at room temperature. The abundant electrons from NPs can boost AD and biogas yield via inducing hydrolysis and acidification stages. The nZVI (0.5 g/L) and Fe_3O_4 NPs (1 g/L) have generated mesophilic biogas effectively after 80 days, which produced 140.34 and 137.13 liters of biogas, respectively (Zhang et al., 2019a, 2019b).
- Adding different concentrations of iron oxide NPs (0.32 g/L) and further NPs doped with granular activated carbon (4 g/L) using food waste as a substrate permitted them to utilize abundant VFA in AD. Under mesophilic conditions, though, VFA buildup is harmful and, due to the maximum protein degradation, leads to the ammonia production. For instance, at mesophilic and thermophilic temperatures, IO-NPs-doped granular activated carbon has produced the desirable biogas yield by improving robust microbial interactions, suggesting the significance of IO-NPs (Juntupally et al., 2022).
- The COD removal, TS, and VS% values of NPs are accountable in producing the biogas yield. In a study, the different TS and VS% values were observed for Co, Ni, Fe, and FeO NPs. Adding nZVI (0.1%) and iron powder (1.6%) to sewage sludge can boost AD, resulting in desirable methane output by 25.2% and 40.8%, respectively (Suanon et al., 2017).
- For obtaining a better COD removal performance with an enhanced methane yield, proper optimization of the microbial community is also an important aspect. In this regard, researchers incorporated citrate-coated Fe_3O_4 NPs during AD for the efficient bioremediation

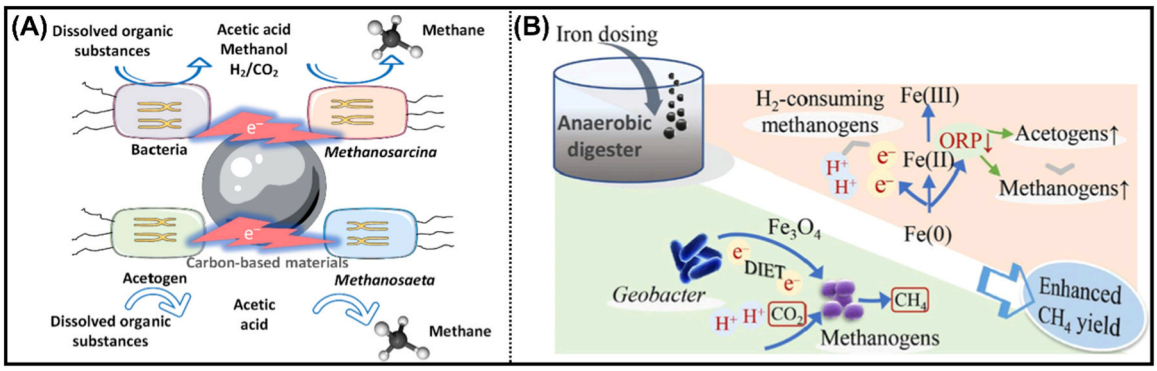

Fig. 9. (A) Potential mechanism of the robust chains of DIET established by carbon-based materials in anaerobic digester. (B) Mechanism of iron-based materials in anaerobic digester. Adapted from (Liang et al., 2022; Tian et al., 2020).

of crude glycerol from biodiesel production into biogas (49.8%) (Boscaro et al., 2022).

- Among Co and Ni NPs, the concentration-dependent biogas and methane yield can be observed. At optimal concentrations of Co NPs (1.0 ppm), a substantial increase in the biogas (1.64) and methane yields (1.86-fold). Similarly, for Ni NPs at concentrations of 2.0 ppm, a significant rise in biogas (1.74-fold) and methane (2.01-fold) yields was observed (Abdelsalam et al., 2017).
- In AD systems, iron bioavailability naturally fluctuates between 1.7% and 3.1%, potentially leading to lower Fe^{2+} levels. Previous studies signified that the acute phase prevention effect of Fe_2O_3 NPs depended on concentrations of NPs and exposure time, while an excess of Fe_2O_3 NP concentrations drastically inhibited the methane yield (Ünşar et al., 2018). These findings infer that elevated NP concentrations in an anaerobic digester during the HRT may disrupt the steady state of AD.

4.4. Graphene and its allied materials

Unfortunately, most research has been focused on finding the best GO doses, and still there is minimal information about the effect of multiple operating variables regarding GO-incorporated AD systems. Furthermore, researchers did not hypothesize the consequences of GO associated with the major stages of AD. In addition, microbial interactions of GO on DIET at different stages of AD are not clear. Thus, thorough and state-of-the-art research is needed for understanding the application-oriented mechanisms of GO in AD.

Using conductive graphene in the AD alters the syntrophic interactions between bacteria that produce acid and archaea, which finally lead to generating methane. In this manner, the exponential proliferation of different bacteria and archaea will occur via DIET during methanogenesis (Lin et al., 2018).

Recently, researchers have studied numerous nanoengineered grazynes as potential CH_4/CO_2 membranes, which improve biogas quality through selectivity, steady flow, and increased CO_2 separation per gram (Viñes et al., 2023). Grazynes are graphynes with stripes of graphene and can connect carbon atoms with sp^2 hybridization through acetylenic links.

- More importantly, at optimal concentrations, trace elements—cobalt (Co), nickel (Ni), and iron (Fe)—in graphene materials can improve methanogenesis by inducing enzyme activity and maximizing biogas yield in anaerobic digesters (Abdelsalam and Samer, 2019).
- Significantly, nano-engineered grazyne membranes can upgrade biogas in contrast to that of transition metal compounds and MXenes due to their lightweight nature.
- Grazyne membranes filter better than activated carbon and can absorb CO_2 quickly, reaching hundreds of mmol/g values in nanoseconds, while the activated carbon-based materials barely reach a few mmol/g after hours of exposure. The superior selectivity makes grazyne-based materials one of the alternatives for improving the biogas quality.
- Researchers have been designing various graphene materials as conveniently flexible and tuned to increase the chemical separation and attain an optimal methane concentration (95% v/v) needed for the commercialization of biogas.

4.5. Bio-carbon and Iron-based conductive materials

Conductive materials strengthen bacteria-methanogenic archaea bonds and allow them to use e-pili for transferring electrons to methanogens. Evidently, activated carbon and biochar can regulate the AcoD using food waste and MOSW and inducing DIET and methanogenesis (Fig. 9A). Earlier studies showed that in lab-scale semi-continuous reactors, activated carbon has a large surface area that helps anaerobic microbes grow well (Zhang et al., 2022, 2017; Wang et al., 2020). Meanwhile, Ma et al. synthesized activated carbon particles (61–75 μm), which considerably boosted methane emission by 22%. The micron--sized activated carbon should have mitigating effects on the methane generation due to the increased substrate binding capacity.

The iron-based CMs like magnetite (Fe_3O_4), hematite (Fe_2O_3), and ZVI (Fe^0) are accountable for boosting the AcoD performance and methane production (Tian and Yu, 2020; Ma et al., 2020).

- Iron-based CMs can promote methanogenesis via iron-microorganism synergy (Fig. 9B). AcoD, which employs food waste and sewage sludge, produces 1.24 times more methane from Fe^0 than Fe_3O_4 (Liang et al., 2021b). Other research studies have observed that replacing nano-size Fe^0 with Fe_3O_4 increased methane generation (Kassab et al., 2020). Basically, nanoscale Fe^0 enhances substrate biodegradability and enzymatic activity, boosting microbial growth, and further, nanoscale Fe_3O_4 oxidizes and reduces electrons between methanogens and syntrophic bacteria effectively (Jin et al., 2019; Zhao et al., 2021). All these studies yielded technically different results, which were mainly based on physiochemical and metabolic effects of iron-based CMs.

  The researchers also investigated the multifarious effects of Fe^0 and Fe_3O_4 on AD, among them, Fe_3O_4 has greatly increased AD; however, Fe^0 had no peculiar impact.
- The coenzyme F_{420} activity is very crucial for inducing the methanogenesis in the AD. In this regard, Fe^0 NPs display superior coenzyme (F_{420}) activity as compared with that of Fe_3O_4 NPs. For example, *Syntrophomonas* was more prevalent than in the Fe^0 system, and Fe_3O_4 enhanced its growth (Liang et al., 2022).

  Significantly, carbon- and iron-based CMs have an influential role in the microbial heterogeneity of AcoD and potentiate desirable biogas yields. Previous research studies also emphasized that AcoD can be promoted using activated carbon, food waste, and sewage

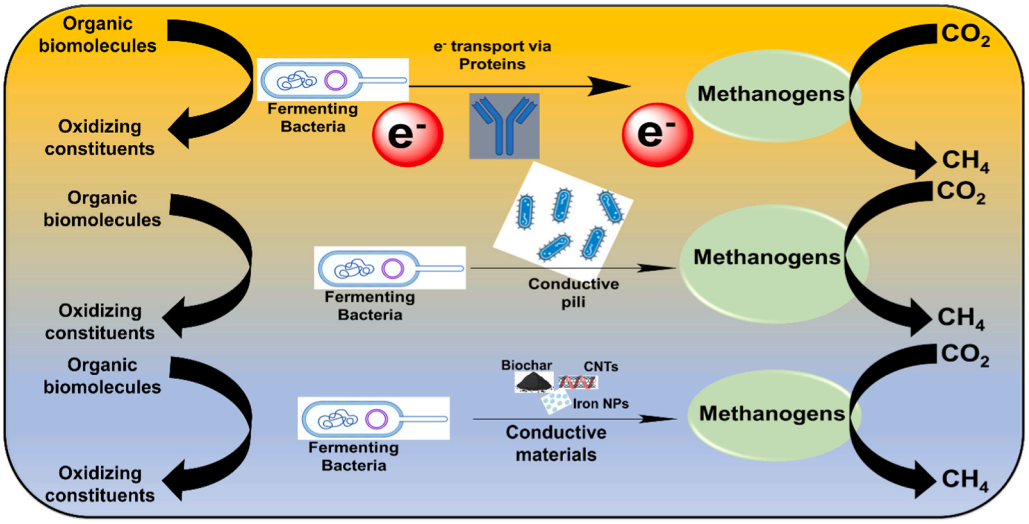

Fig. 10. Schematic illustration of the fermenting bacteria involving the methane formation via ET, e-pili, and CMs through the oxidation of different organic biomolecules.

sludge as waste substrates via microbial heterogeneity (Wang et al., 2020; Zhao et al., 2021).

- Liang's research group focused on the application of CMs by employing AcoD of food waste and sewage sludge substrates, which positively modify microbial population heterogeneity. For example, EPS-degrading microbes like *Mesotoga* and *Thermovirga* have shown influential effects after the addition of Fe^0 as compared with that of charcoal (Liang et al., 2022). In another study, the ZVI may have increased solubilization and hydrolysis, and thereby boosting methanogenesis (Thapa et al., 2024; Kumar et al., 2021; Deng et al., 2022; Ko et al., 2018).

To promote AD, researchers used various carbon-based CMs; granular activated carbon, carbon fiber, and biochar are conveniently attractive for high methane-generating potentials.

- Biochar acting as a physicochemically superior additive, offering high specific surface area, excellent porosity, oxygen-containing functional groups, high cation exchange capacity, and strong electrical conductivity. Using forestry waste for biochar production has reduced GHG emissions to 2.74 kg CO_2-eq/kg. Further, its biocompatible nature has abundantly contributed to freshwater eutrophication and fossil depletion.
- The electrical conductivity is crucial for obtaining high methane yield in anaerobic digesters, which depends on several features such as quantity of biochar and concentrations of buffers and different ions such as sulfides, metal ions, and carbonates. Under the significant influence of microbial proliferation, biochar releases abundant inorganic ions from substrates and superiorly decomposes the organic matter. Thus, incorporation of the biochar is proportionately affecting the superior electrical conductivity in many anaerobic digesters.
- The biochar is majorly associated with four key mechanisms in AD as follows: (i) amplification of functional microorganisms; (ii) the DIET; (iii) augmentation of refractory chemical breakdown; and (iv) stabilization of the reactor. The elevated specific surface area and microporous architecture of biochar create an optimal environment for microorganisms to establish colonies and enhance macro- and micronutrient content via sorption. These macro- and micronutrients include phosphorus, nitrogen, calcium, potassium, sodium, iron, and magnesium, which signify intense metabolic activities of syntrophic bacteria (*Geobacter metallireducens*) and methanogenic archaea (*Methanosarcina barkeri*) (Devi and Eskicioglu, 2024).
- The microbial community study of DIET has reported an improvement in AD utilizing biochar by enhancing co-cultures, formate, and H_2 transfer channels across microbial species, and consequently increased the CH_4 generation (Kumar et al., 2021). Noteworthy, CMs can efficiently facilitate EET and stimulate abundant e-pili, thereby supporting fermentative bacteria in generating methane through the oxidation of organic macromolecules (Fig. 10) (Barua and Dhar, 2017).

From earlier studies, it has been suggested that the optimal concentration of impurities like H_2S in biogas can be as low as 50–10,000 ppm based on the type of material fed into the digester. Any alterations from these concentrations can corrode the engine and metal parts through induced H_2S/SO_2 emissions from combustion, thus, it is better to remove H_2S.

- For instance, a case study was conducted in Thailand at a small pig farm for the biogas-to-electricity system, where 2% potassium iodide (KI) incorporated with activated carbon was employed as an H_2S adsorbent and achieved a 100% H_2S removal efficiency with an adsorption capacity of 0.062 kg of H_2S per kilogram of adsorbent (Pipatmanomai et al., 2009).
- While in Palestine, biogas production is significantly constrained, with only one operational industrial-scale biogas facility. The cost burden for the CMs like activated carbon to remove H_2S is a highly bottlenecked issue, and thus, currently researchers are focusing on maximum yield biomass precursor and the suitable impregnation reagent (Sawalha et al., 2020). Among different biomass wastes such as coffee grounds, eucalyptus bark, and nut shells, the eucalyptus bark has produced superior carbonization yield around 35%. Based on these discussions, it is known that Palestine and other poor countries use eucalyptus bark in combination with KOH to achieve superior adsorption capacity and beneficial H_2S removal efficiency from biogas.

In this context, the type of effective biomass precursor and the physicochemical characteristics (i.e., particle size and dosage), as well as the economic burden of CMs, are practically crucial to assess the economic feasibility for large-scale industrial applications.

Biochar enhances the DIET process among microbes by transporting electrons to eliminate total organic carbons, and thereby regulates AD. The particle size of biochar mainly affects microbial colonization and methanogen proliferation, which sophisticate methane emissions (Qiu

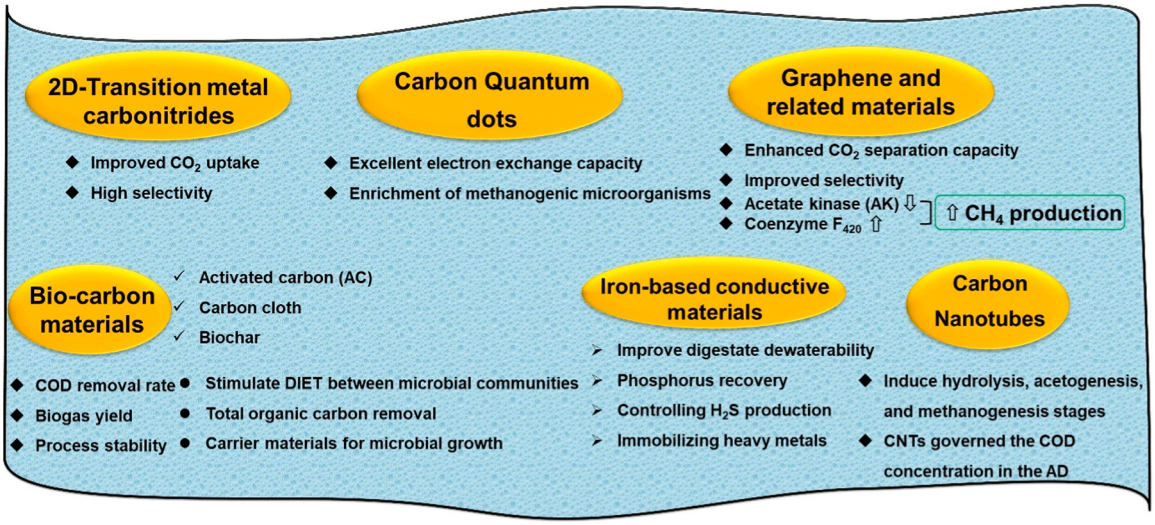

Fig. 11. Overview depicting the role of different materials and their characteristic features for the biogas production.

et al., 2019; Kumar et al., 2021; Luo et al., 2015; Lü et al., 2019). In another study, Zhang et al. pyrolyzed wood waste at 750°C for 60 min, to synthesize biochar, where its oxygen-rich composition and induced graphitization resulting in high methane yield (21.5%) (Zhang et al., 2022).

In this manner, understanding of miscellaneous case studies is significantly crucial, which reveals the insights of biochar on the ecological and economic effects in the biogas sector.

- In a recent case study in Singapore, the economic aspects of incorporating biochar to AD systems in biogas plants were investigated. The significant economic profit may be achieved with the addition of biochar, as demonstrated by the annual methane productivity of 1.01×10^8 m^3 . The operational and administrative expenses of the biogas yield facility ranged from USD 4.1–6.1 million, whereas the revenue from electricity generated by biogas combustion was around USD 10 million (Zhang et al., 2020).
- Another case study in California has determined the environmental application of biochar using digestate may store 110 MMT of CO_2 emissions in soil (Breunig et al., 2019). The biochar fertilization in soil might provide up to 2.74 t CO_2 eq t^{-1} and can remarkably diminish N_2O/CH_4 emissions and transform the soil into a carbon sink (Khan et al., 2013).

These studies are crucial for understanding the ecological and commercial perspectives of different CMs in the biogas sector. Further research on various material applications is necessary to achieve sophisticated economic stability in the biogas field.

Based on the above-mentioned discussions, a brief summary of characteristic properties of 0–2D materials for biogas generation was depicted in Fig. 11.

5. Machine learning models in the biogas production

Machine learning algorithms typically examine data from AD systems to enhance efficiency and performance through predictions. When choosing an ML model, it is crucial to evaluate the specific objectives and limitations of the model, together with the quality of data and the complexity of the problems being tackled. These include forecasting biogas yield, methane concentration, energy output, etc.

The optimized ML-based approaches can produce strong predictive models for thorough understanding of crucial parameters in the design of AD. Proper ML models can conveniently execute a planning approach for anaerobic digesters in the utilization of different waste. To obtain the maximum biogas yield, feedstock's properties (i.e., C/N and VS/TS%) should be monitored initially. Further, set the two inputs as constants and allow ML models to find the best values for operational conditions from the collected data sets. In this way, biogas plants can reduce the cost burden and minimize the time delay by speeding up the AD (Li et al., 2022a).

For instance, the estimation of feature significance analysis after ML model building can determine the critical workflow of operating factors that affected the biogas yield and primarily emphasizes that alkalinity and VFAs are the key factors that influencing the effectiveness of AD (Yildirim and Ozkaya, 2023).

- For the convenient optimization of miscellaneous ML models to achieve the success rate in the AD, the following parameters must be considered: (a) process efficiency, (b) cost burden, (c) reliability, (d) safety, and (e) data-driven forecasting. Meanwhile, for optimizing productivity, the meticulously monitored key operating parameters are as follows: (i) substrate characteristics, (ii) OLR, (iii) pH, (iv) HRT, (v) solids retention time, (vi) TS% and VS%, (vii) temperature, (viii) alkalinity, and (ix) VFA.
- Multifarious ML approaches were investigated to predict the effectiveness of AD through operational biogas data over the year (Yildirim and Ozkaya, 2023). To solve the problems with AD systems, biogas facilities can use advanced ML models and Industry 4.0 technologies to make the systems more efficient and make on-site control easier.

To understand the latest trends regarding the divergent ML models, the current review article proposed the keyword co-occurrence analysis utilizing different search criteria observed from the last decade (2015–2025) using "Web of science." In the first set, "machine learning," "anaerobic digestion," "biogas," and "waste" are the keywords for searching the data. Herein, total of 103 documents and 587 keywords were identified, among which 34 keywords were met the threshold. Further, in the other set, "nanoparticles," "anaerobic digestion," "biogas," "pretreatment," and "waste" are the searching keywords. Here, a total of 67 documents and 389 keywords were identified, and a threshold of 43 keywords was observed. Based on these observations, the data was plotted using VOS viewer software (Fig. 12). From these data analyses, it is inferred that numerous publications utilized ML techniques.

Furthermore, typical ML models such as artificial neural network (ANN), genetic algorithm (GA), support vector machine (SVM), and random forest (RF) were analyzed in a detailed manner. The main

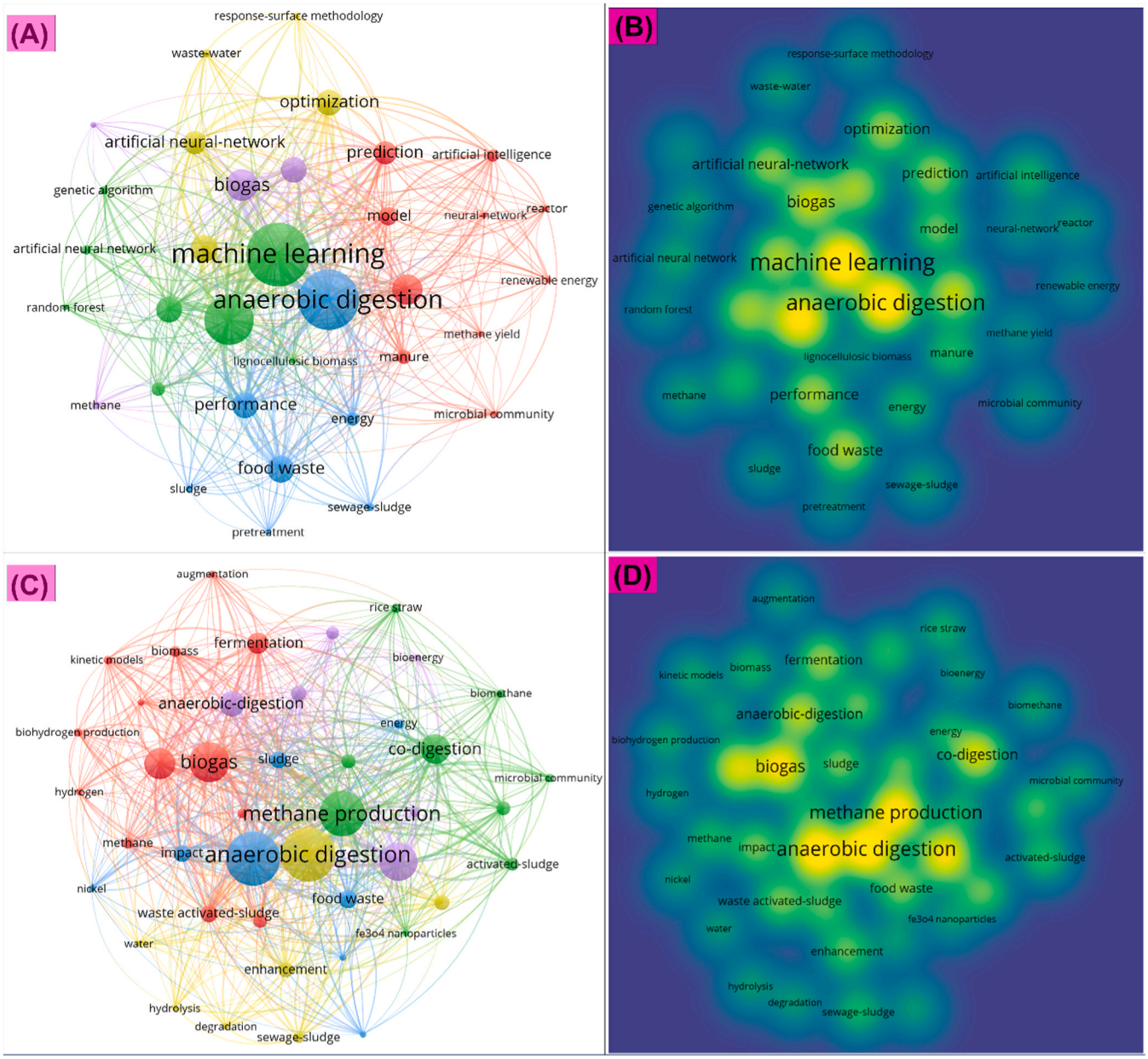

Fig. 12. Keyword co-occurrence analysis for biogas production using ML models; (A) network and (B) density visualization plots. Keyword co-occurrence analysis in biogas production related to nanoparticles. (C) network and (D) density visualization plots.

keywords are "anaerobic digestion," and "biogas," along with each of the ML models (Fig. 13). The bibliographic data is as follows: (i) For ANN: total documents (202) and keywords (970) were identified, among which keywords (81) met the threshold; (ii) For GA: total documents (59) and keywords (392) were identified, with threshold keywords (25); (iii) For SVM: total documents (36) and keywords (267) were identified, with threshold keywords (11); and (iv) For RF: total documents (27) and keywords (211) were identified, with threshold keywords (9).

Based on the above-mentioned bibliographic data, ANN is the most exploited ML model regarding the biogas estimation in the last decade.

- In the near future, combining algorithms such as ANN and GA should be a top priority, as those significantly enhance data-driven intelligence for improved modeling, forecasting, and decision-making throughout the AD (Barik and Murugan, 2015).
- A constructive summary was presented for comparative investigations of ML models and AD system linkages (Clercq et al., 2019a; Andrade et al., 2022). Significantly, ANNs are the leading ML models and are utilized in multiple AD systems.

Recently, researchers conducted the pilot study in 273 and 113 AD facilities in the USA and Germany, respectively. Some key factors affecting AD efficiency were provided based on the ML models, including data envelope analysis and stochastic gradient boosting. The leading technological factors that affected German projects include (i) pre-treatment stages; (ii) box-type technology in digesters; and (iii) co-digestion of different waste sources. Meanwhile, primary features influencing AD efficiency in USA projects are (i) the type of covered lagoon digester, (ii) co-digestion of various waste types, and (iii) the type of mixed plug flow digester. These pilot study results are useful for selecting the best type of AD system to use in large-scale operations.

The main research gap involved in the above pilot study includes the unavailability of comprehensive panel data; only a snapshot of typical operational conditions in biogas facilities in Germany and the US is

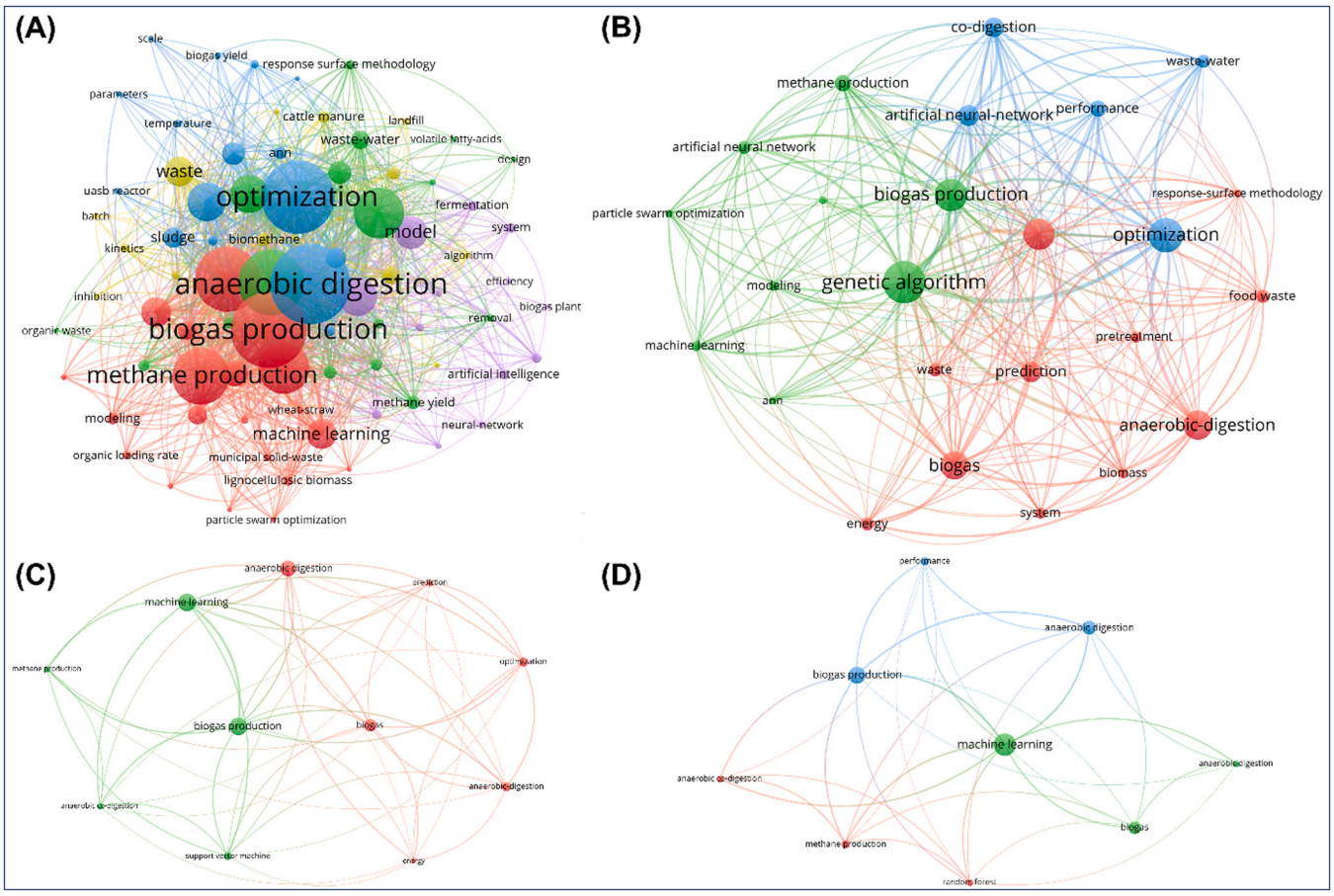

Fig. 13. Keyword co-occurrence analysis for biogas production using different ML models including (A) ANN, (B) GA, (C) SVM, and (D) RF, respectively.

available. However, the data was limited due to the inadequate specific features regarding the reactor working, including (i) OLR, (ii) HRT, (iii) temperature, and (iv) digestate recycling. These are critical characteristics that governments must apply during the selection of new projects related to AD facilities (Clercq et al., 2019b).

6. Limitations, research needs, and future perspectives

Generally, limited methane production and reactor instability plague single-substrate AD systems, i.e., WAS. Landfilling and incineration hurt the environment and economy; hence, sustainable food waste disposal is essential (Kanellos et al., 2024). The substrates' C/N ratio was slightly below optimal, increasing ammonia buildup and AD instability (Tiong et al., 2023). A limited pH range may overlook the synthesis of VFA and the microbial activity occurring at very high pH levels.

- In-depth research investigations are urgently required to properly explore a broader pH range, microbial interactions, persistent operating conditions, scalability, and the impact of VFA development on AD. For example, proper monitoring of distinguished characteristics, such as alkalinity, nutritional equilibrium, and reduction of ammonia-free inhibitors, is essential for AD maintenance (Harun et al., 2019).
- Concerning Fe_3O_4 NPs, researchers should concentrate on highly negatively charged Fe_3O_4 NPs for large-scale industrial uses. These NPs can efficiently produce methane in AD, but in excess, Fe_3O_4 NPs cause selective mortality of microbes. Unfortunately, excessive concentrations of Fe_3O_4 NPs could hinder further utility in the above-discussed applications, and thus Fe_3O_4 NPs are not much explored in thermophilic biogas generation units (Boscaro et al., 2022).

For multiple reasons, AD systems use biomass-based carbon materials sparingly, but unfortunately, biomass ash and trace elements like Na^+, Mg^{2+}, S^{2-}, K^+, and Ca^{2+} detrimentally affect the AD. For instance, materials synthesized from the lignocellulose can pollute the environment by emitting heavy metals and organic pollutants. Still more, divergent digestates exert adverse potentials on the ecosystem, and thus it is crucial to validate these materials for the safety and stability of the AD system (Abbas et al., 2021). Significantly, the ammonium-free nature of biochar can adversely affect the AD due to its ash content and trace elements. In this regard, the production of biodiesel urgently requires pretreated feedstock and recyclable high-yielding catalysts (Lü et al., 2019; Nguyen et al., 2024; Zhen et al., 2023).

Both physicochemical and biological pretreatments are still suffering from having few limitations as follows:

- For instance, SE pretreatment requires high operational pressure to operate the AD in different substrates, which procures the higher cost burden.
- Again, due to the formation of inhibitory compounds during AD, substrates suffer from the reduced enzymatic hydrolysis, which adversely affects the biogas yields.
- While, in the case of SCE pretreatment in AD digestors, sophisticated safety cabinet setups need to be standardized to cope with high energy demand.
- Continuous monitoring of CO_2 explosion is highly challenging to meet industrial-level optimization.
- The longer duration of SCE pretreatment obstructs the smooth operation and is quite similar to that of the thermo-biological pretreatment, which requires extended hours.

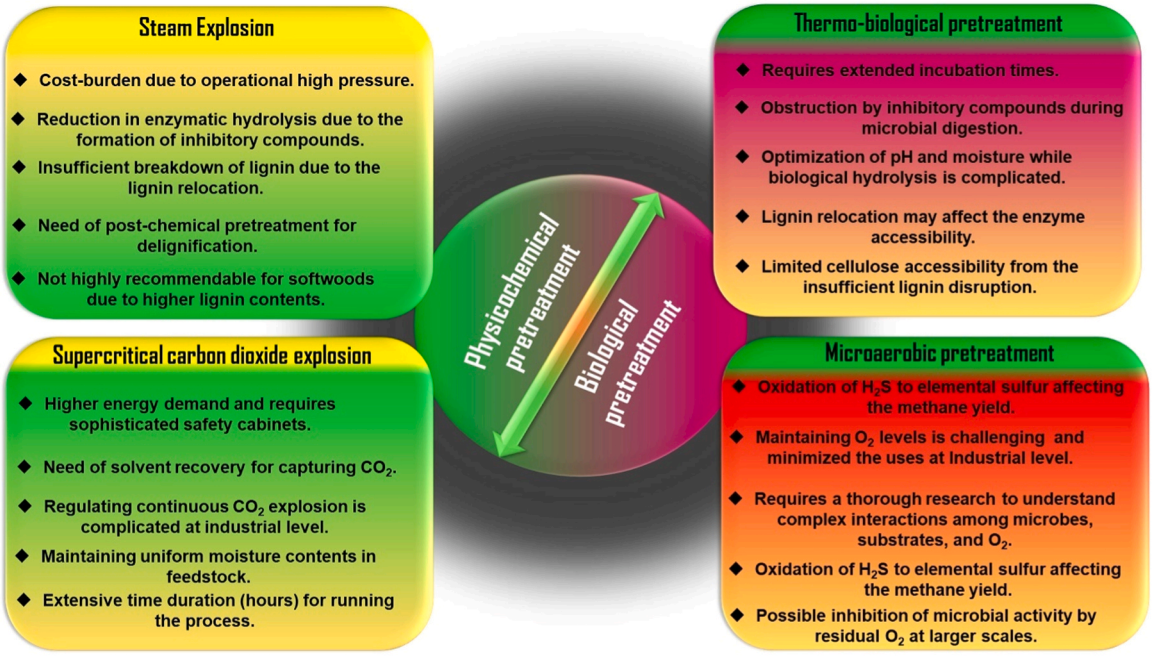

Fig. 14. Summary of limitations for the physicochemical (SE and SCE) and biological (thermo-biological and microaerobic) pretreatments.

- More importantly, regulating operational parameters such as pH and moisture is highly difficult in the thermo-biological pretreatment.
- Thermo-biological pretreatment led to the insufficient lignin disruption, which consequently inhibited the cellulose accessibility from the substrates during AD process.
- During microaeration, there are higher possibilities for the early oxidation of H_2S to elemental sulfur, which decreases the methane production.
- The convenient maintenance of O_2 levels is not comfortable and affects the methane yield at industrial and large-scale settings.

Fig. 14. summarizes the possible limitations of physicochemical and biological pretreatments.

Unfortunately, the ML models have limited ability to predict the AD and biogas yield because of numerous operational parameters, such as OLR, pH, hydraulic and solids retention times, and substrate types. The combined use of multiple ML models is significant for precise prediction of the biogas yield by regulating several operating conditions in the AD. Therefore, more actionable pathways need to be considered for rectifying the existing limitations of ML models as follows:

- In this regard, the combination of ANN and GA models has successfully controlled multiple input and output parameters; (i) the duration of AD, (ii) an optimal C/N ratio, (iii) pH, and (iv) the biogas yield.
- The convenient control of nonlinear relationships and guess variables in complex structures during AD is crucial. In this way, an adaptive neural-fuzzy inference system (ANFIS) is a novel model designed based on fuzzy systems and neural networks, and moreover, it provides superior adaptability to input parameters, system, and operating settings. Currently, several studies are relying on its exploitation for the prediction of biogas yield.
- So far, ML technologies haven't observed the main groups of bacteria and archaea that affect how well AD works. Hence, the proper incorporation of genetic information and key operational characteristics into the dataset can develop a unique model for the accurate prediction of the biogas yield.
- The synthetic minority over-sampling technique (SMOTE) recently makes new data samples within the original data set using the K-nearest neighbors (KNN) method to balance the data and reduce issues caused by uneven data distribution. Through combining SMOTE, RF, and GA models, scientists can enhance the ability to learn from imbalanced data and improve the accuracy of the regression model (Wang et al., 2024a, 2024b).
- The technological content, input waste materials, and output products in various biogas plants might differ significantly. Such variation indicates the challenge of creating a universal ML software application appropriate to resolving operational constraints across various projects. Future studies should concentrate on pinpointing the predominant operational issues encountered by many projects globally to advance the creation of an open-source, general-purpose ML graphical user interface software tool.
- Governments, biogas project investors, and industrial corporations should advocate for the adoption of such instruments. Numerous biogas initiatives globally rely on public financing to support operations, making it essential that financial resources designated for biogas facilities are utilized effectively.
- The machine learning models can add more value to large-scale AD by (i) making it easier to combine all the data in AcoD facilities, (ii) allowing sensors to measure important process parameters in real time, and (iii) creating models based on a set of features that can be easily judged in later feature importance and partial dependence analyses.

In AD settings, especially external factors are interacting adversely in complex ways, which affect the understanding of the workflow process. In a case study conducted in Austria, researchers developed a unique ML architecture known as the Temporal Fusion Transformer (TFT) for forecasting biogas production rates, which presents many advantages: (i) it speeds up the sluggish adaptation of the AD in altered operating circumstances, (ii) it manages both categorical and continuous data via a comprehensive set of input variables, and (iii) an intrinsic attention mechanism enables it to delineate the specific issues. This approach is more beneficial in biogas plant operators to deal with the decision problems of black box models like ANNs (Sappl et al., 2023).

Regarding the developing countries such as India, biogas projects face numerous challenges as follows: (i) high start-up costs; (ii) timely delivery of the necessary feedstocks for either mono- or co-digestion at

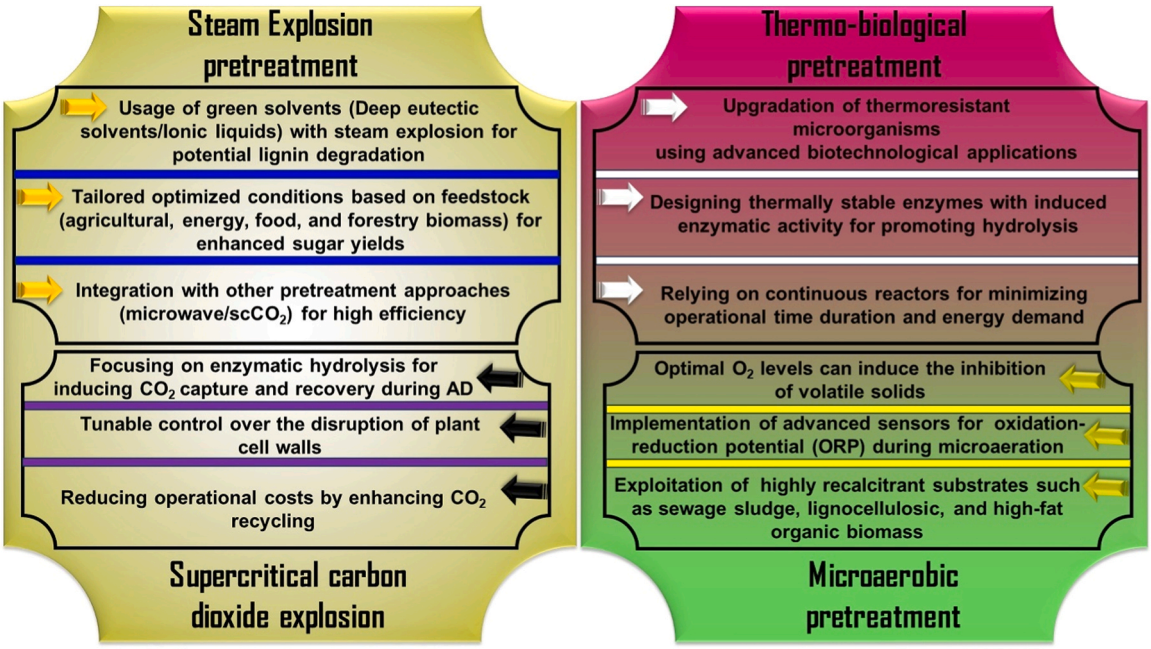

Fig. 15. Overview of future prospects for the physicochemical (SE and SCE) and biological (thermo-biological and microaerobic) pretreatments.

all biogas facilities nationwide; (iii) proper collection of agricultural waste due to inadequate tools; (iv) incorrect sorting of municipal solid waste at the source; and (v) the need for an abundance of water in biogas plants. To cope with these challenges, these countries need a stringent governmental policy and substantial private investments. Moreover, due to the escalating costs of petroleum products and cooking gas, transitioning to biomethane for energy requirements is a more economical and sustainable alternative. In this context, a temporal evaluation of waste availability, particularly agricultural wastes that are seasonal and accessible just post-harvest, is necessary to formulate biogas feedstock supply chains and optimize storage planning (Dey and Thomson, 2022; Position and Planet, 2018; Godoy et al., 2024). Future work is required on strategies to incentivize source segregation by residents for municipal solid waste.

- In the SE pretreatment, green solvents such as DESs and ionic liquids are highly useful for the substantial degradation of lignin constituents.
- Feedstocks such as agricultural, energy, food, and forestry biomass should be thoroughly optimized to enhance sufficient glucose formation.
- Combination of SE pretreatment with other pretreatments such as microwave and SCE are also beneficial for achieving high efficiency in the biogas production.
- During the AD process, careful monitoring should be prioritized on enzymatic hydrolysis of substrate, which is beneficial for sufficient capturing and recovery of CO_2 in the SCE pretreatment.
- While in thermo-biological pretreatment, thermoresistant microorganisms need to be upgraded via employing biotechnological applications.
- More precisely, thermally stable enzymes can play a key significant role in promoting the enzymatic activity by inducing hydrolysis.
- The optimal maintenance of O_2 levels is complicated, and therefore, skillful sensors must be implemented, which can conveniently determine ORP values in the microaerobic pretreatment.
- By regulating optimal O_2 levels in the digestion of substrates, significant proportions of volatile solids can be inhibited.
- Significantly, highly recalcitrant substrates, including lignocellulose, sewage sludge, and more fat-based organic biomass, should be thoroughly exploited through microaerobic pretreatment, which can produce substantial biogas.

Fig. 15. summarizes the future prospects of physicochemical and biological pretreatments in a graphical representation for a better understanding.

7. Conclusions

This review explored recent advances, research gaps, and future perspectives of pretreatment strategies, anaerobic co-digestion, and different materials involved in biogas production. Based on the thorough review analysis from previous reports, it is suggested that a combination of two or more pretreatments is more beneficial for achieving high biogas yield with minimal limitations. Furthermore, this review signifies different mechanisms involved in pretreatments with regard to their applications. The comparative bibliographic study and keyword co-occurrence analysis on the ML models were thoroughly investigated. The statistical results were strongly emphasized using the frequency of articles dealing with SDGs. Significant case studies in the biogas sector at the lab, pilot scale, and industrial level with their ecological and economical perspectives were discussed. Typical mechanisms and characteristics of various materials from CQDs to CMs were explained. Still, there are key research gaps in the progress of MXene, graphene, and CNT-based materials for the biogas sector that need to be focused on more in the near future.

CRediT authorship contribution statement

Roopkumar Sangubotla: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Gangaraju Gedda:** Writing – review & editing. **Jongsung Kim:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Myung-Geol Pang:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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